

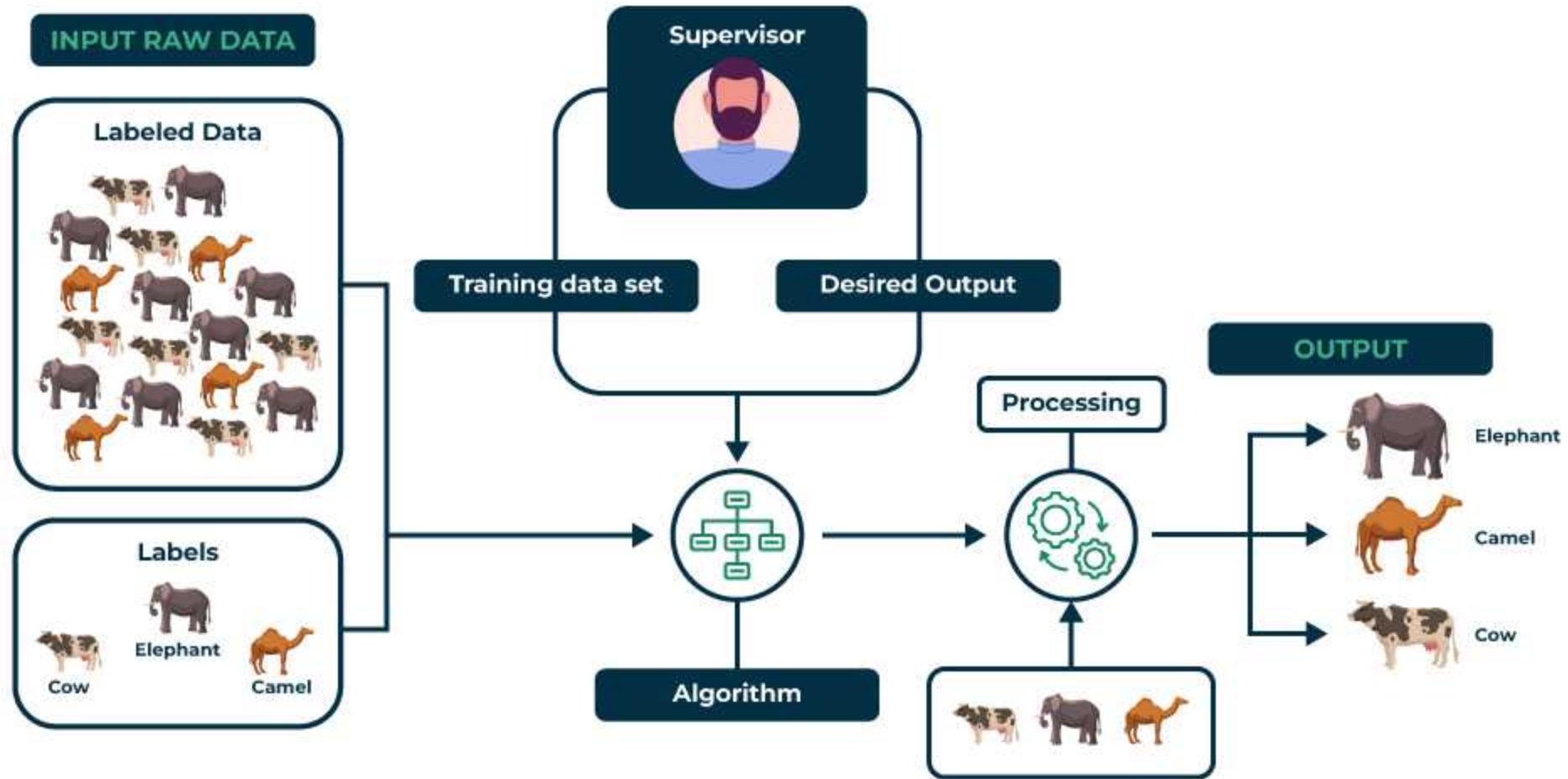
Machine Learning



- ◇ Chapter VI:
- ◇ Supervised Learning: Part I

◇ Prof. Dr. Aderval S. Luna

Supervised Learning



Supervised Learning

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graph TD; A[Supervised Learning] --> B[Regression]; A --> C[Classification];
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The diagram illustrates the structure of Supervised Learning. At the top, a blue rectangular box contains the text 'Supervised Learning'. Two arrows originate from the bottom of this box, pointing downwards to two separate ovals. The left oval is green and contains the text 'Regression'. The right oval is light blue and contains the text 'Classification'.

Regression

Classification

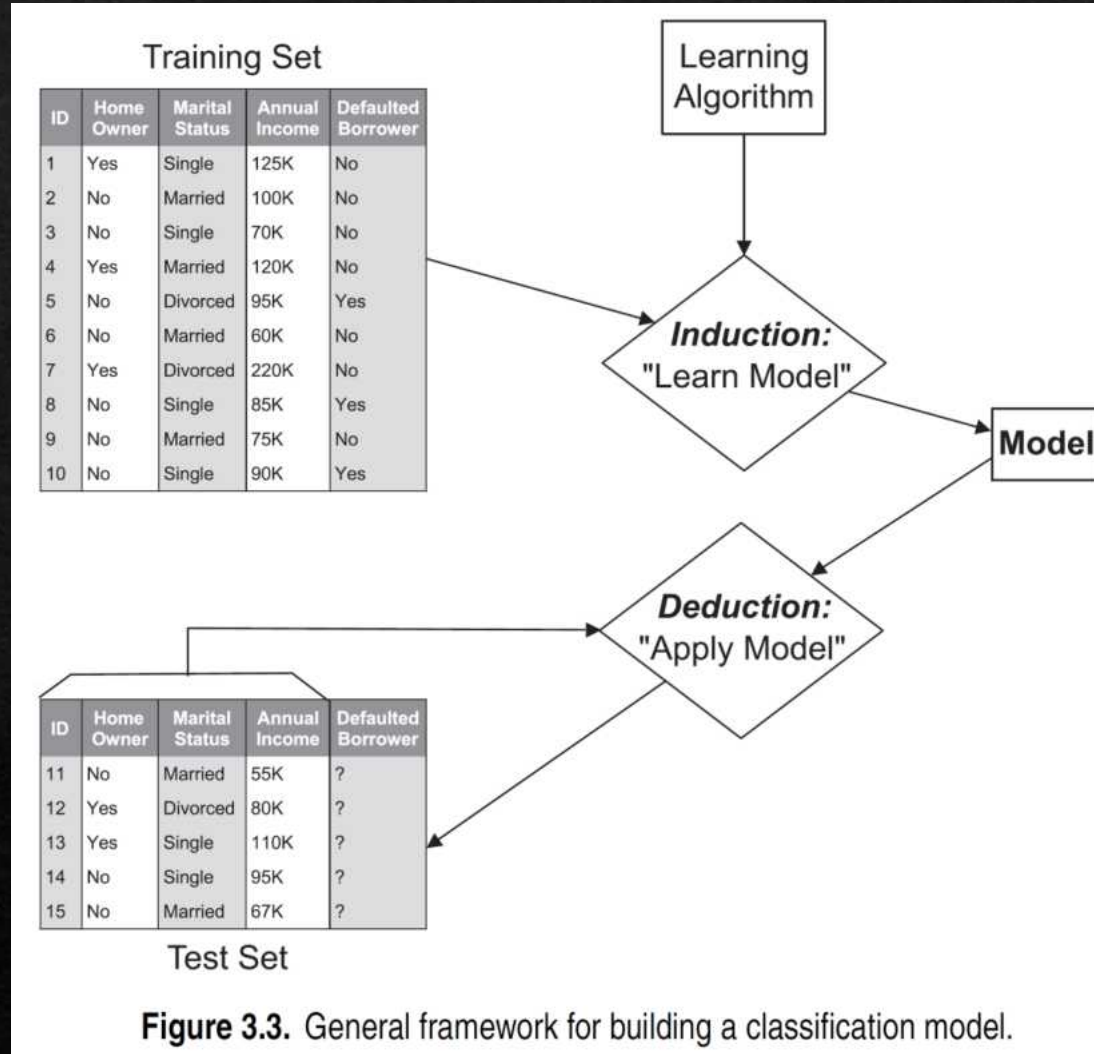
Classification: Definition

- Given a collection of records (training set)
 - Each record is by characterized by a tuple $(\mathbf{x}, y)$, where $\mathbf{x}$ is the attribute set and y is the class label
 - ◆ $\mathbf{x}$: attribute, predictor, independent variable, input
 - ◆ y : class, response, dependent variable, output
- Task:
 - Learn a model that maps each attribute set $\mathbf{x}$ into one of the predefined class labels y

Examples of Classification Task

Task	Attribute set, x	Class label, y
Categorizing email messages	Features extracted from email message header and content	spam or non-spam
Identifying tumor cells	Features extracted from x-rays or MRI scans	malignant or benign cells
Cataloging galaxies	Features extracted from telescope images	Elliptical, spiral, or irregular-shaped galaxies

General Approach for Building Classification Model



Supervised Learning Algorithms

◆ Classification

- ◆ Linear Discriminant Analysis, LDA
- ◆ Quadratic Discriminant Analysis, QDA
- ◆ Mixture Discriminant Analysis, MDA
- ◆ Regularized Discriminant Analysis, RDA
- ◆ Partial Least Squares Discriminant Analysis, PLS-DA

◆ Classification

- ◆ Orthogonal Partial Least Squares Discriminant Analysis, OPLS-DA
- ◆ Self Independent Modeling of Class Analogy, SIMCA
- ◆ Logistic Regression
- ◆ Bayes Classifier
- ◆ Rule-Based Classifier
- ◆ Decision Trees, DT

Classification Methods: Part 1

- ◆ Linear Discriminant Analysis, LDA
- ◆ Quadratic Discriminant Analysis, QDA
- ◆ Mixture Discriminant Analysis, MDA
- ◆ Regularized Discriminant Analysis, RDA
- ◆ Partial Least Squares Discriminant Analysis, PLS-DA
- ◆ Orthogonal Partial Least Squares Discriminant Analysis, OPLS-DA
- ◆ Self Independent Modeling of Class Analogy, SIMCA
- ◆ Logistic Regression
- ◆ Bayes Classifier
- ◆ Rule-Based Classifier
- ◆ Decision Trees, DT

Linear Discriminant Analysis, LDA

- ◆ LDA is the oldest classification method, developed by R. Fisher, and published (1936): Annals of Eugenics 7(2): 179 -188.
- ◆ The method is designed to enable separation into two classes.
- ◆ The training set consists of two matrices $\mathbf{X}_1$ and $\mathbf{X}_2$, which have I_1 and I_2 rows, respectively.
- ◆ The initial assumptions are as follows:
- ◆ Each class ($k = 1$ or 2) is modeled by the normal distribution $N(\mu_k, \Sigma_k)$
- ◆ Covariance matrices of the two classes are identical $\Sigma_1 = \Sigma_2 = \Sigma$

Linear Discriminant Analysis, LDA

- ◆ The LDA classification rule is very simple and states that a new sample $\mathbf{x}$ belongs to the class which is closer by the Mahalanobis metric.
- ◆ $d_k = (\mathbf{x} - \boldsymbol{\mu}_k) \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}_k)^t$
- ◆ In practice, the unknown expectations $\boldsymbol{\mu}_k$ and the covariance matrix $\boldsymbol{\Sigma}$ are replaced by their estimators:
- ◆ $\mathbf{m}_k = \left(\frac{1}{I_k} \right) \sum_{i=1}^{I_k} \mathbf{x}_i$
- ◆ $\mathbf{S} = \left(\frac{1}{I_1 + I_2 - 2} \right) (\tilde{\mathbf{X}}_1^t \tilde{\mathbf{X}}_1 + \tilde{\mathbf{X}}_2^t \tilde{\mathbf{X}}_2)$
- ◆ In this formula, $\tilde{\mathbf{X}}_k$ is the centered matrix $\mathbf{X}_k$.

Linear Discriminant Analysis, LDA

- ◆ $d_k = (\mathbf{x} - \boldsymbol{\mu}_k)\boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu}_k)^t$
- ◆ Setting $d_1 = d_2$, we obtain an equation of a curve, which separates the classes.
Here, the quadratic terms $\mathbf{x}\mathbf{S}^{-1}\mathbf{x}^t$ are reduced and the equation becomes linear.
- ◆ $\mathbf{x}\mathbf{w}_1^t - v_1 = \mathbf{x}\mathbf{w}_2^t - v_2$ (The terms at two sides are called LDA-scores)
- ◆ Where
- ◆ $\mathbf{w}_k = \mathbf{m}_k\mathbf{S}^{-1}$ and $v_k = 0.5\mathbf{m}_k\mathbf{S}^{-1}\mathbf{m}_k^t$
- ◆ $f_1 = \mathbf{x}\mathbf{w}_1^t - v_1$ and $f_2 = \mathbf{x}\mathbf{w}_2^t - v_2$
- ◆ An object belongs to Class 1, if $f_1 > f_2$ and, vice-versa, to Class 2, if $f_1 < f_2$

Linear Discriminant Analysis, LDA

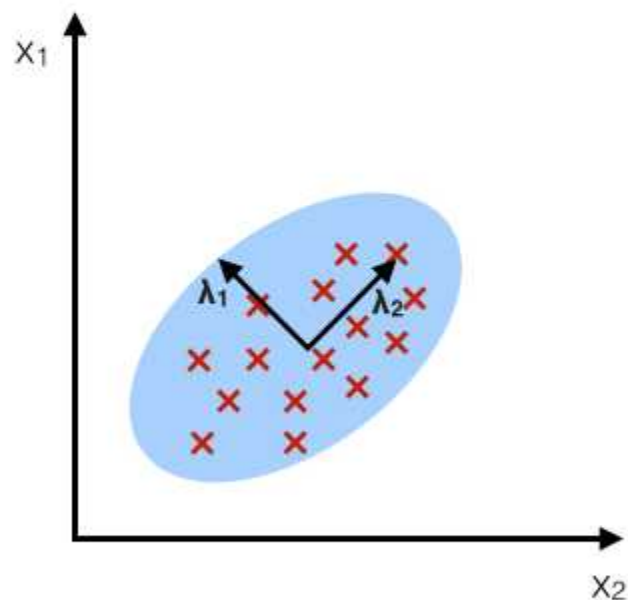
- ◆ The main LDA difficulty is the inversion of matrix S .
- ◆ If S is degenerate, the method cannot be used.
- ◆ Therefore, before using LDA, the original matrix X is often replaced by the PCA-scores matrix T , which is not singular.

Linear Discriminant Analysis, LDA

- ◆ Linear Discriminant Analysis can be used to perform supervised dimensionality reduction, by projecting the input data to a linear subspace consisting of the directions which maximize the separation between classes.
- ◆ The dimension of the output is necessarily less than the number of classes, so this is in general a rather strong dimensionality reduction, and only makes sense in a multiclass setting.
- ◆ LDA is a special case of QDA (Quadratic Discriminant Analysis), where the Gaussian for each class are assumed to share the same covariance matrix: $\Sigma_k = \Sigma$ for all k .

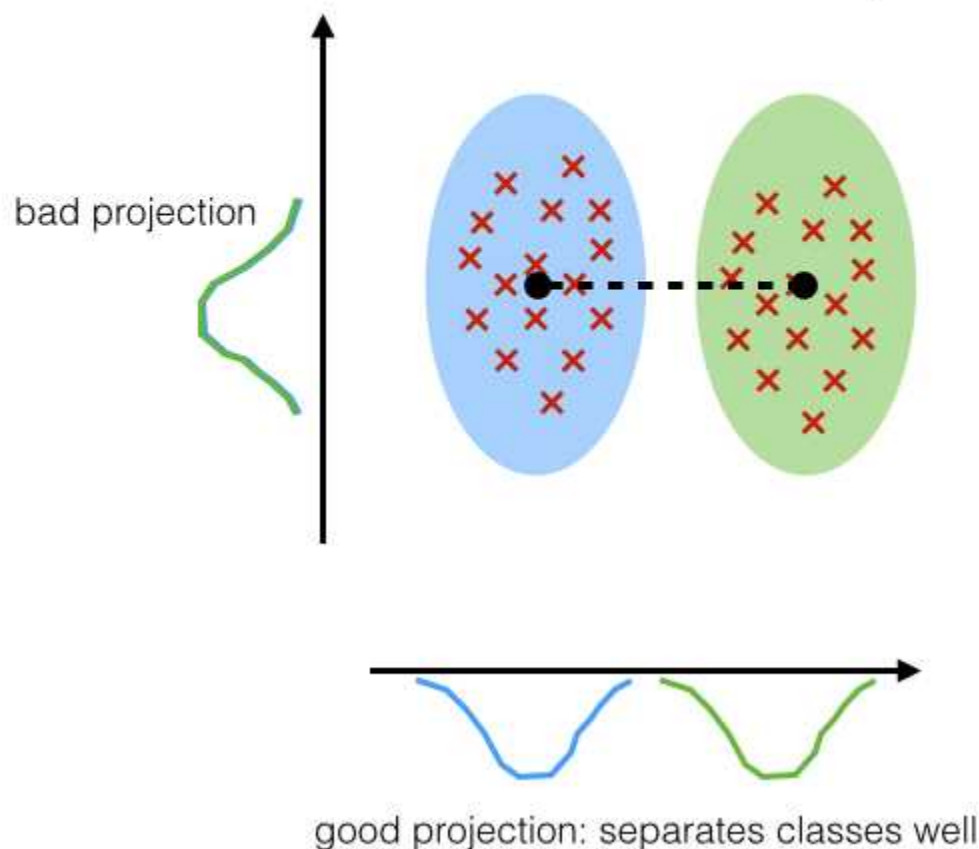
PCA:

component axes that maximize the variance

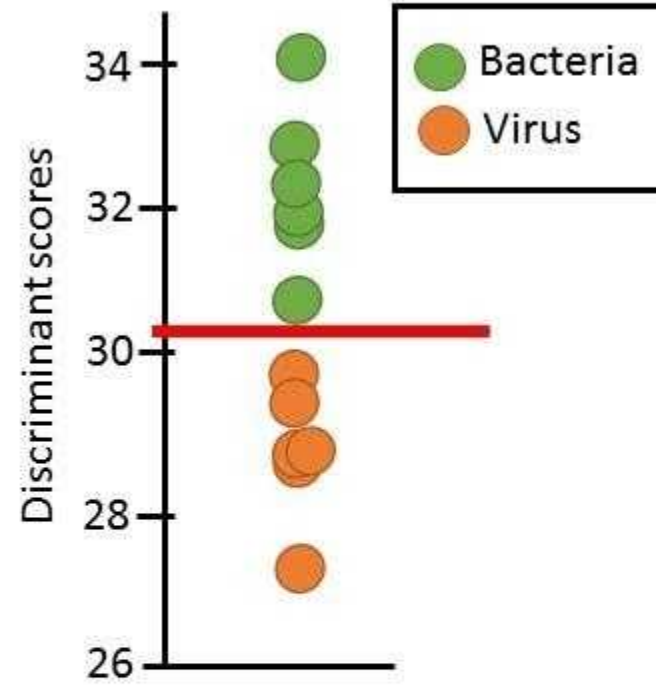
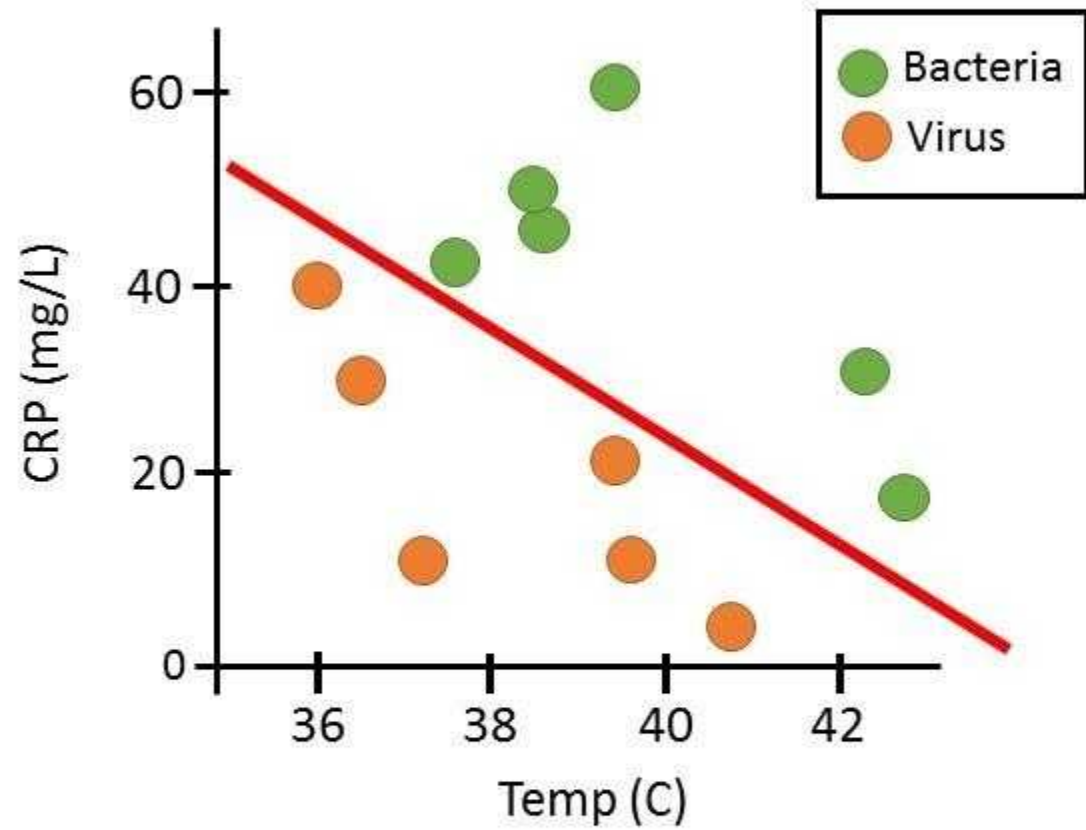


LDA:

maximizing the component axes for class-separation

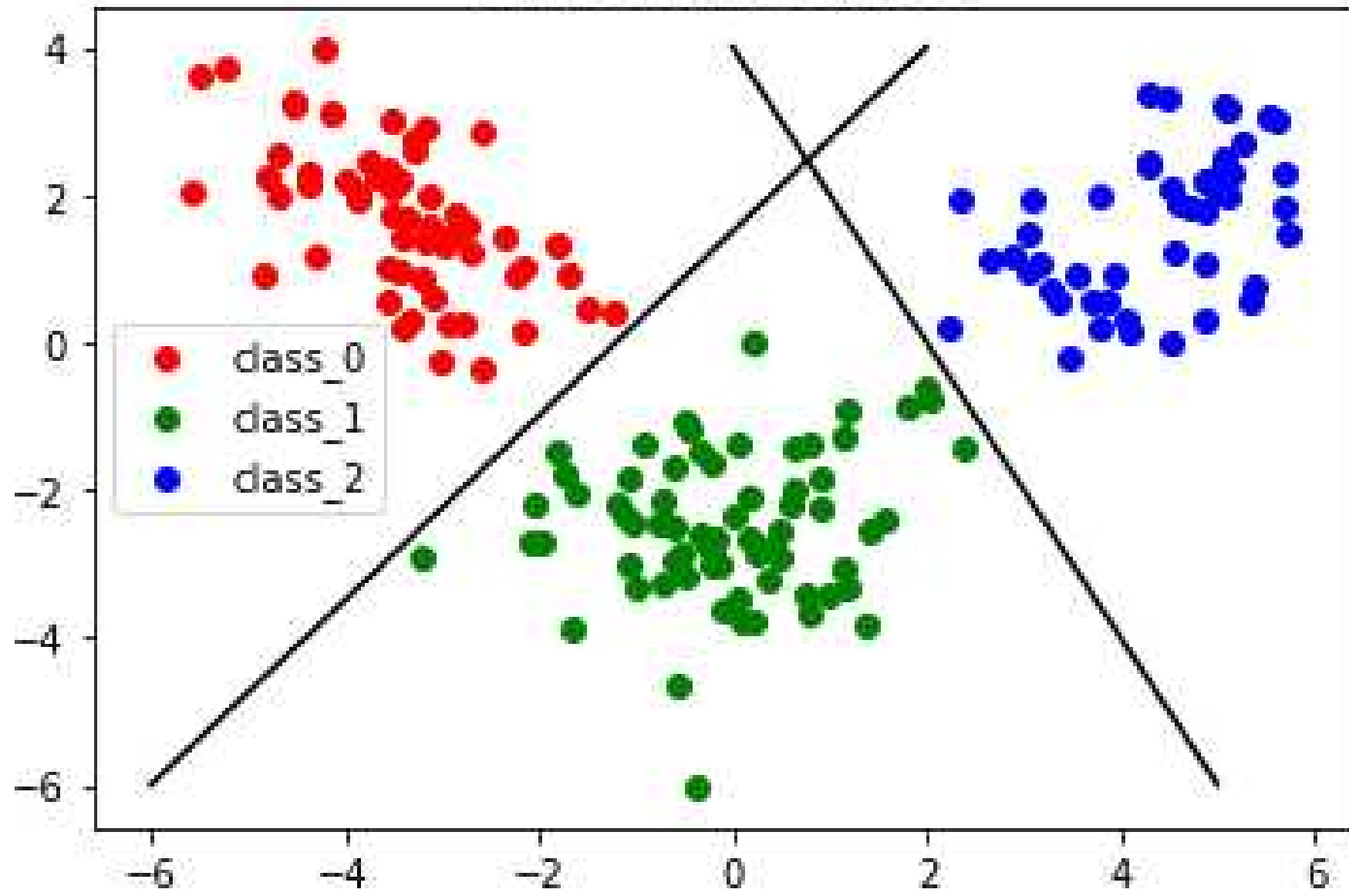


LDA



TileStats

LDA of WINE dataset



Advantages and Disadvantages of LDA

◆ Advantages of LDA:

- ◆ LDA is easy to apply.

◆ “Drawbacks of LDA:”

- ◆ LDA does not work when the covariance matrix is singular, for a large number of variables.
- ◆ It requires regularization, for example, by PCA.
- ◆ The LDA method is unsuitable if the covariance matrices of the classes differ.

Quadratic Discriminant Analysis, QDA

- ◊ QDA is a generalization of LDA.
- ◊ QDA is a multiclass method that can be used for simultaneous classification of several classes.

Quadratic Discriminant Analysis, QDA

- ◆ A training set consists of K matrices $\mathbf{X}_1, \dots, \mathbf{X}_K$, which have $I_1, \dots, I_K$ rows (samples) correspondingly.
- ◆ The number of variables (columns) is the same for all matrices and equals J .
- ◆ The initial assumptions are as follows:
- ◆ Each class is modeled by a normal distribution $N(\mu_k, \Sigma_k)$.
- ◆ The covariance matrices Σ_k are **different**.

Quadratic Discriminant Analysis, QDA

- ◆ Then the QDA scores are calculated by:
- ◆ $f_x = (\mathbf{x} - \boldsymbol{\mu}_k) \boldsymbol{\Sigma}_k^{-1} (\mathbf{x} - \boldsymbol{\mu}_k)^t + \log \left(\det(\boldsymbol{\Sigma}_k^{-1}) \right) \quad k = 1, \dots, K$
- ◆ The QDA classification rules is as follows:
- ◆ A new sample $\mathbf{x}$ belongs to the class provides the smallest QDA-score.

Quadratic Discriminant Analysis, QDA

- ◆ Similar to LDA, the unknown expectations μ_k and the covariances matrices Σ_k are replaced by their estimates:
- ◆ $m_k = \frac{1}{I_k} \sum_{i=1}^{I_k} \mathbf{x}_i$ $S_k = \frac{1}{I_k} \tilde{\mathbf{X}}_k^t \tilde{\mathbf{X}}_k$
- ◆ In these formulas, $\tilde{\mathbf{X}}_k$ is the centered matrix $\mathbf{X}_k$.
- ◆ A surface that separates classes k and l is determined by a quadratic equation: $f_k = f_l$

Advantages and Disadvantages of QDA

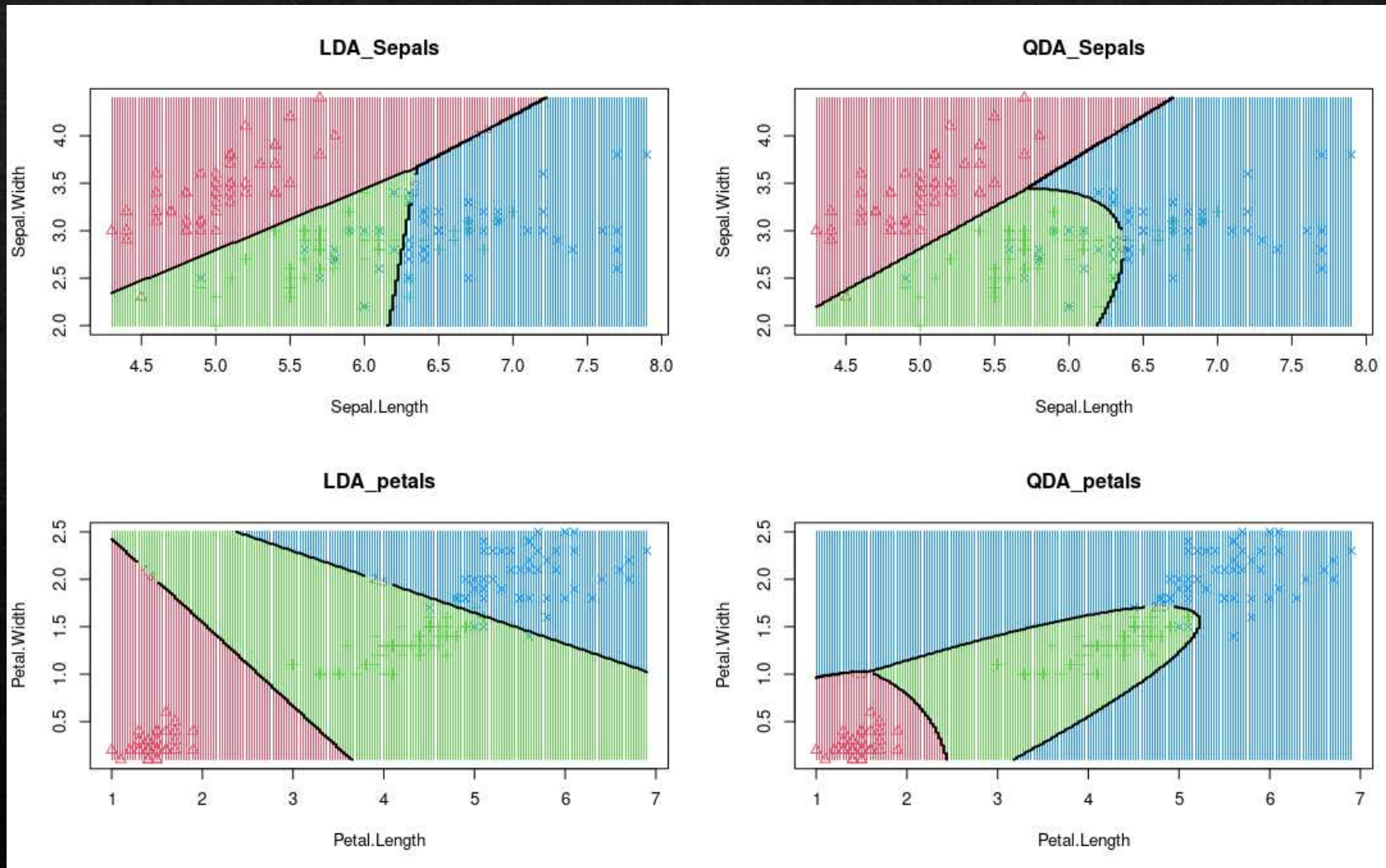
♦ Advantages of QDA:

- ♦ It is suitable when the covariance matrices are different.

♦ “Drawbacks of LDA:”

- ♦ QDA does not work when the covariance matrices are degenerate, for a large number of variables.
- ♦ It requires regularization, for example, by PCA.
- ♦ QDA implicitly uses the assumption of normality.
- ♦ QDA does not allow interchanging the errors of the first and second kind.

Quadratic Discriminant Analysis, QDA

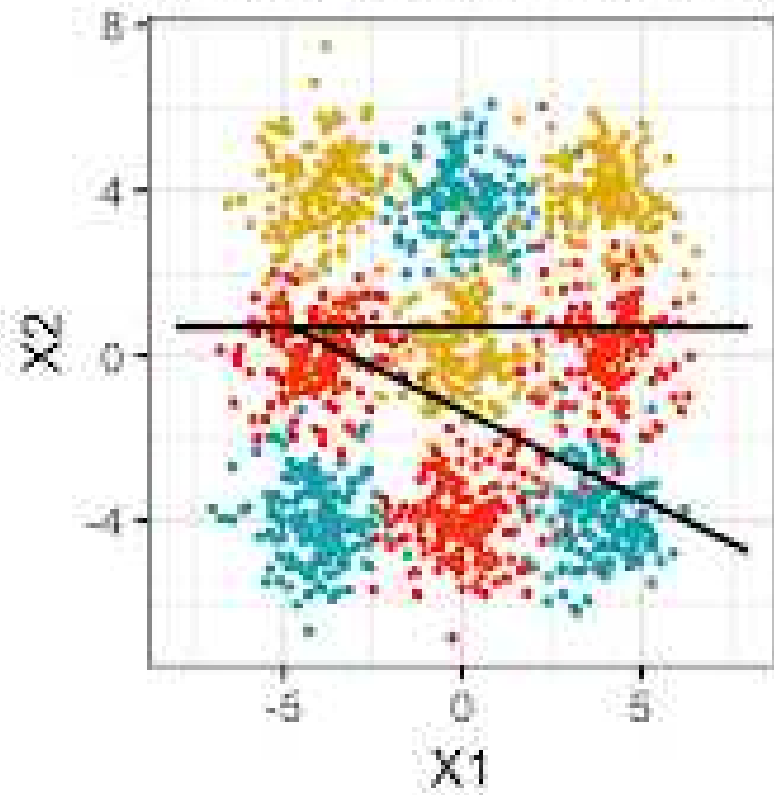


Mixture Discriminant Analysis, MDA

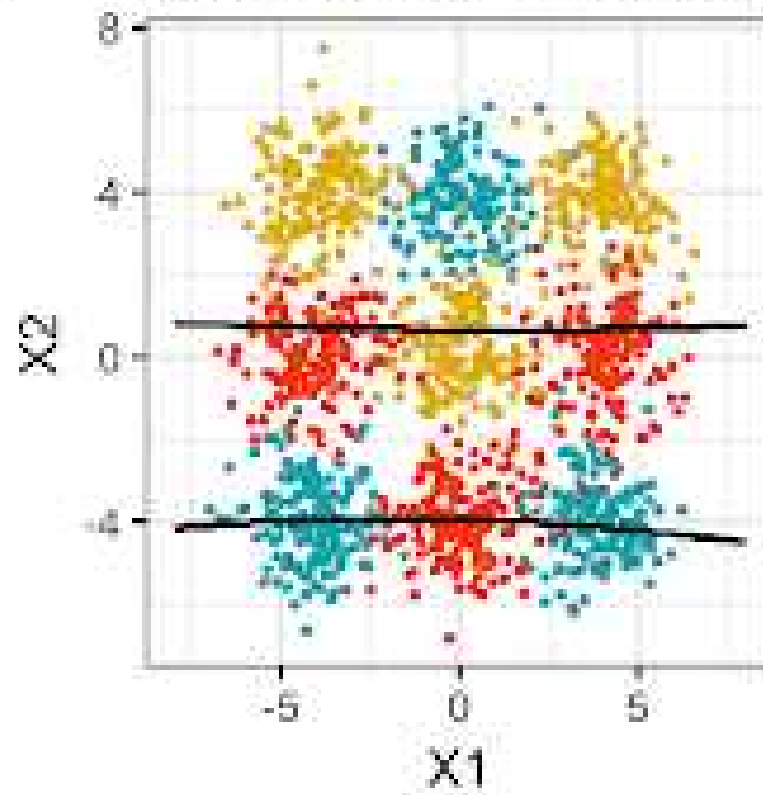
- ◆ Model each class as a mixture of two or more Gaussian components.
- ◆ All components sharing the same covariance matrix.

y • 1 • 2 • 3

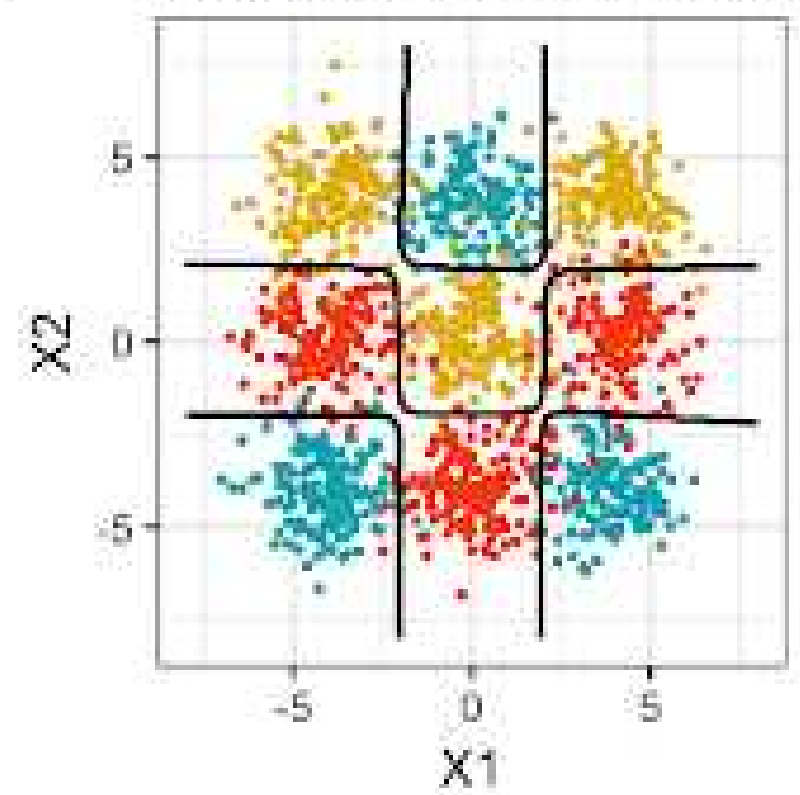
LDA Decision Boundari



QDA Decision Boundar



MDA Decision Boundar



Discriminant Analysis by Gaussian Mixtures FREE

Trevor Hastie , Robert Tibshirani

Journal of the Royal Statistical Society: Series B (Methodological), Volume 58, Issue 1, January 1996, Pages 155–176, <https://doi.org/10.1111/j.2517-6161.1996.tb02073.x>

Published: 05 December 2018



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SUMMARY

Fisher-Rao linear discriminant analysis (LDA) is a valuable tool for multigroup classification. LDA is equivalent to maximum likelihood classification assuming Gaussian distributions for each class. In this paper, we fit Gaussian mixtures to each class to facilitate effective classification in non-normal settings, especially when the classes are clustered. Low dimensional views are an important by-product of LDA—our new techniques inherit this feature. We can control the within-class spread of the subclass centres relative to the between-class spread. Our technique for fitting these models permits a natural blend with nonparametric versions of LDA.

Regularized Discriminant Analysis, RDA

- ◆ Regularized discriminant analysis is a sort of trade-off between LDA and QDA.
- ◆ Remember, in LDA, we assume there is a common covariance matrix for all of the classes.
- ◆ QDA assumes different covariance matrices for all the classes.
- ◆ Regularized discriminant analysis is an intermediate between LDA and QDA, developed by J. Friedman.
- ◆ RDA shrinks the separate covariances of QDA toward a common covariance as in LDA.

Regularized linear discriminant analysis and its application in microarrays ^{FREE}

Yaqian Guo , Trevor Hastie, Robert Tibshirani

Biostatistics, Volume 8, Issue 1, January 2007, Pages 86–100,

<https://doi.org/10.1093/biostatistics/kxj035>

Published: 07 April 2006 **Article history** ▼



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Abstract

In this paper, we introduce a modified version of linear discriminant analysis, called the “shrunk centroids regularized discriminant analysis” (SCRDA). This method generalizes the idea of the “nearest shrunk centroids” (NSC) (Tibshirani *and others*, 2003) into the classical discriminant analysis. The SCRDA method is specially designed for classification problems in high dimension low sample size situations, for example, microarray data. Through both simulated data and real life data, it is shown that this method performs very well in multivariate classification problems, often outperforms the PAM method (using the NSC algorithm) and can be as competitive as the support vector machines classifiers. It is also suitable for feature elimination purpose and can be used as gene selection method. The open source R package for this method (named “rda”) is available on CRAN (<http://www.r-project.org>) for download and testing.

Partial Least Squares Discriminant Analysis, PLS-DA

- ◆ PLS-DA is a variant used when the Y is categorical.
- ◆ PLS is used to find the fundamental relations between two matrices (X and Y), i.e., a latent variable to modeling the covariance structures in these two spaces.

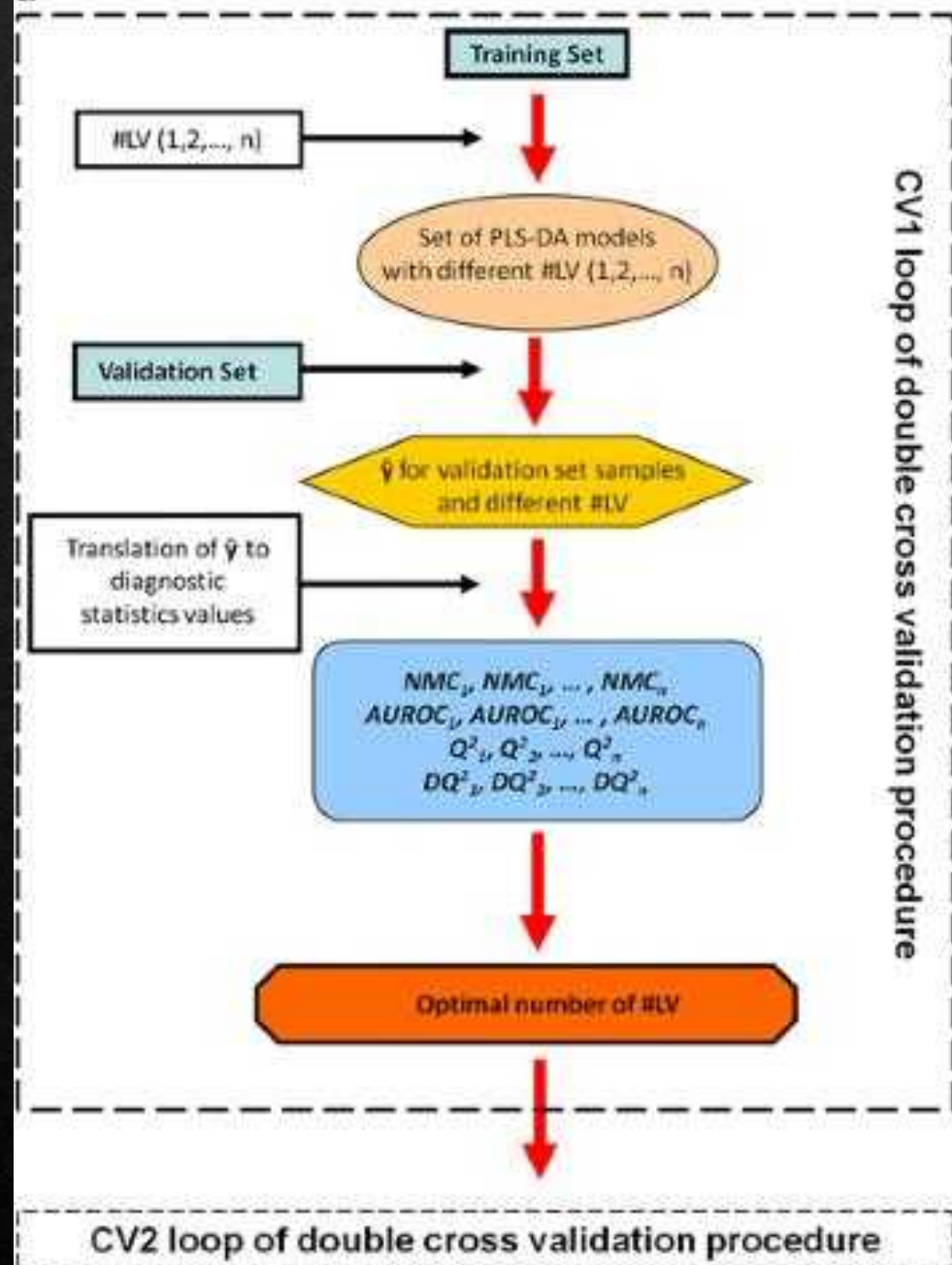
Partial Least Squares Discriminant Analysis, PLS-DA

- ◆ There are two fundamental equations in the PLS-DA model:
- ◆ $\mathbf{X} = \mathbf{T}\mathbf{P}^T + \mathbf{E}$
- ◆ $\mathbf{Y} = \mathbf{T}\mathbf{Q}^T + \mathbf{F}$
- ◆ Now, if we define a $J \times P$ weights matrix $\mathbf{W}$, we can write the scores matrix as
- ◆ $\mathbf{T} = \mathbf{X}\mathbf{W}(\mathbf{P}^T\mathbf{W})^{-1}$
- ◆ and by substituting it into the PLS-DA model, we obtain
- ◆ $\mathbf{Y} = \mathbf{X}\mathbf{W}(\mathbf{P}^T\mathbf{W})^{-1}\mathbf{Q}^T + \mathbf{F}$

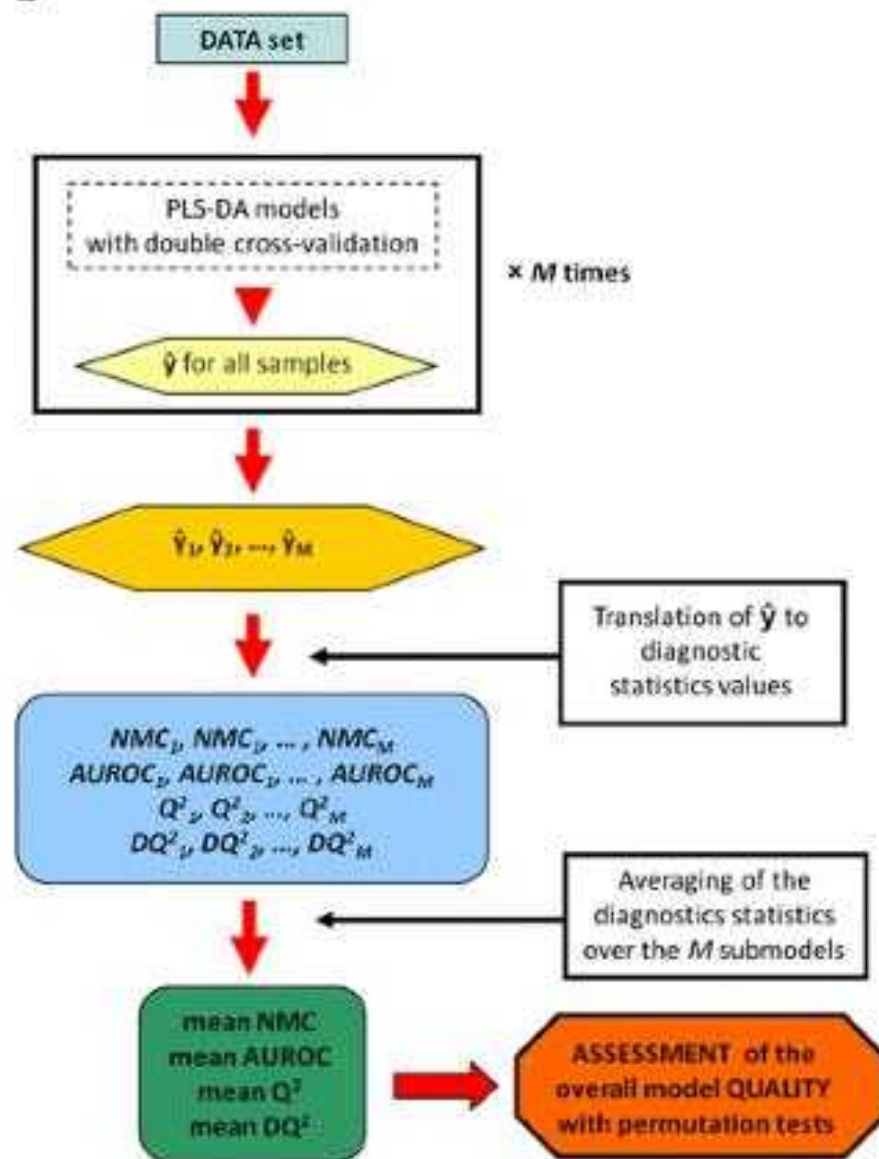
Partial Least Squares Discriminant Analysis, PLS-DA

- ◆ where the coefficient regression vector β is given by:
- ◆ $\hat{\beta} = W(P^T W)^{-1} Q^T$
- ◆ In this way, an unknown sample value of Y can be predicted by $\hat{Y} = B\hat{X}$, i. e.
 $\hat{Y} = XW(P^T W)^{-1} Q^T$
- ◆ Brereton, R. G. & Lloyd, G. R. Partial least squares discriminant analysis: taking the magic way. Journal of Chemometrics 28(4), 213 – 225, 2014.

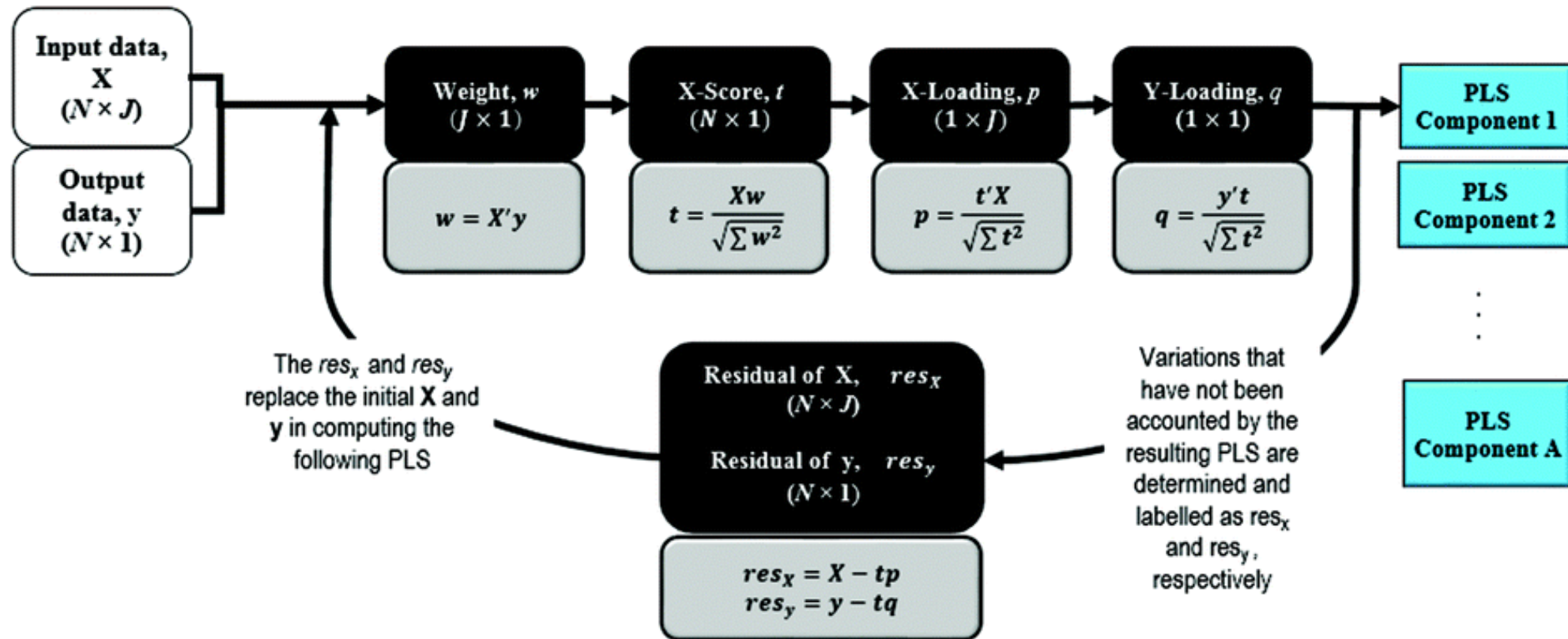
a



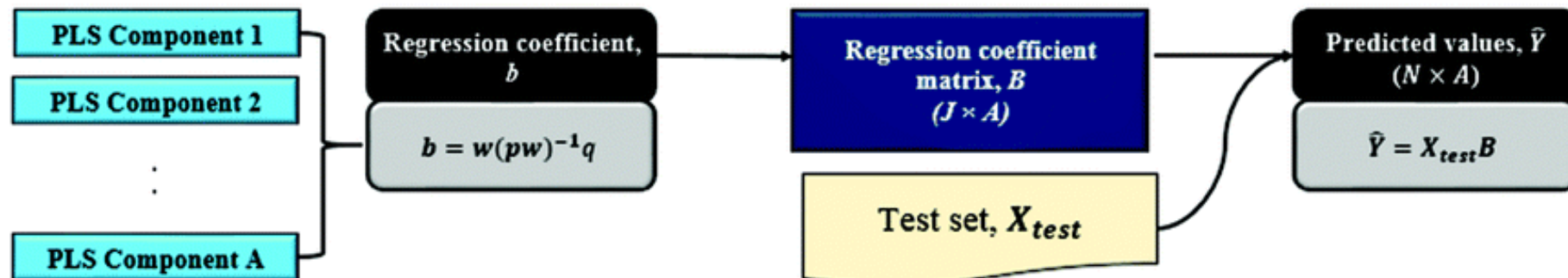
b



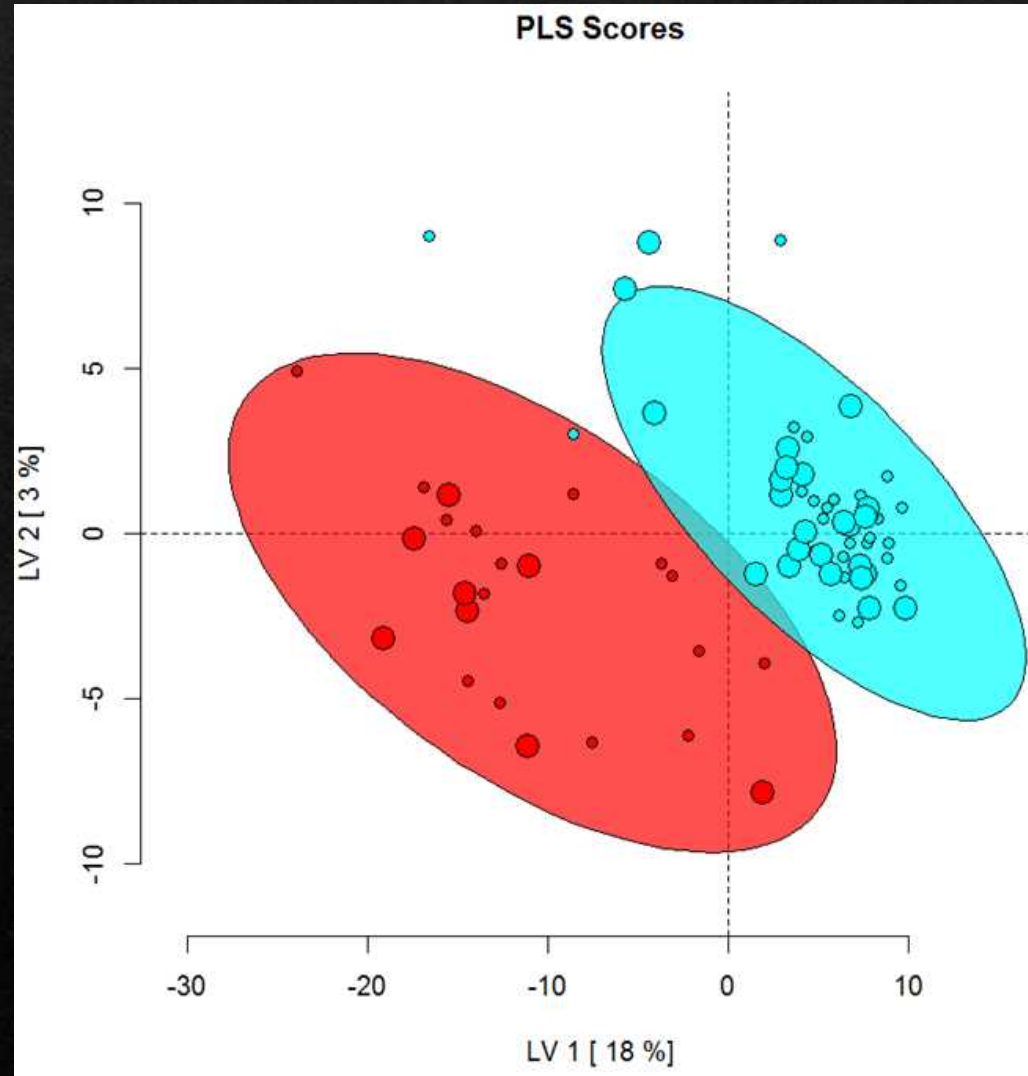
PLS COMPONENT CONSTRUCTION



PREDICTION MODEL CONSTRUCTION



Partial Least Squares Discriminant Analysis, PLS-DA



Advantages and Disadvantages of PLS-DA

◆ Advantages of PLS-DA:

- ◆ It does not use the distribution type assumptions.
- ◆ PLS-DA is applicable to a large number of variables and resistant to the curse of dimensionality.

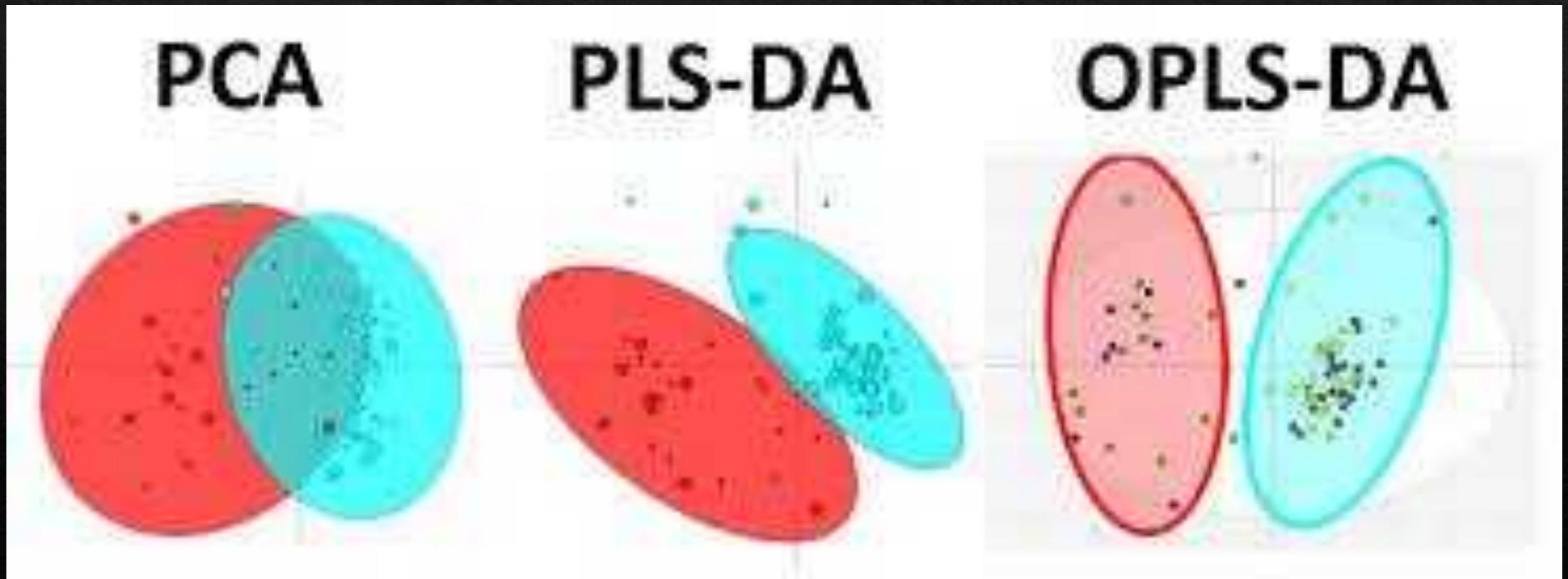
◆ “Drawbacks of PLS-DA:”

- ◆ PLS-DA requires preliminary regression analysis.
- ◆ The result depends on the number of latent variables.
- ◆ PLS-DA is sensitive to outliers.
- ◆ PLS-DA does not work for a small number of samples in the training set.

Orthogonal Partial Least Squares Discriminant Analysis, OPLS-DA

- ♦ Orthogonal partial least squares discriminant analysis (OPLS-DA) was introduced as an improvement of the PLS-DA approach to discriminate two or more groups (classes) using multivariate data.
- ♦ In OPLS-DA, a regression model is constructed between the multivariate data and a response variable that only contains class information.
- ♦ The obvious advantage of OPLS-DA compared with PLS-DA is that a single component serves as a predictor for the class, while the other components describe the variation orthogonal to the first predictive component.

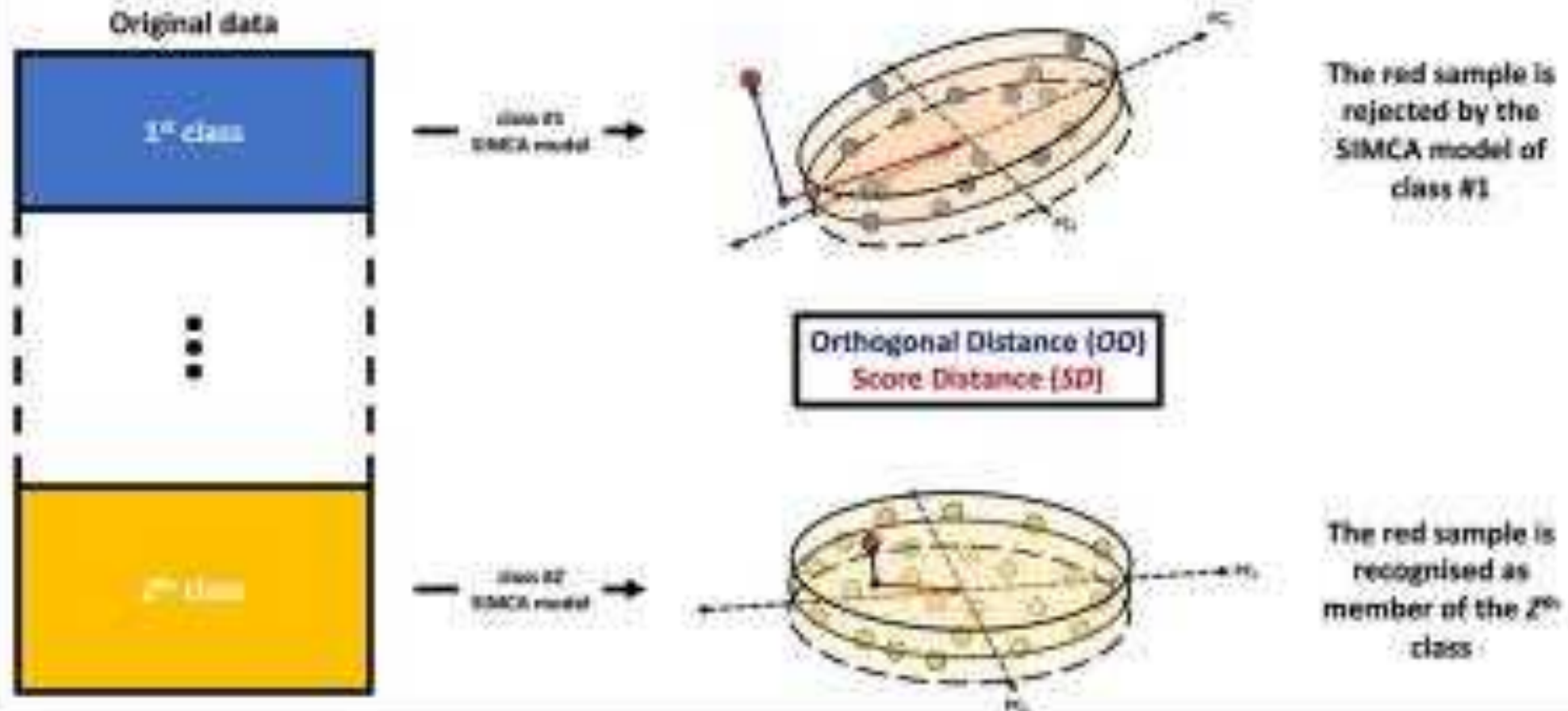
Comparison of PCA, PLS-DA and OPLS-DA



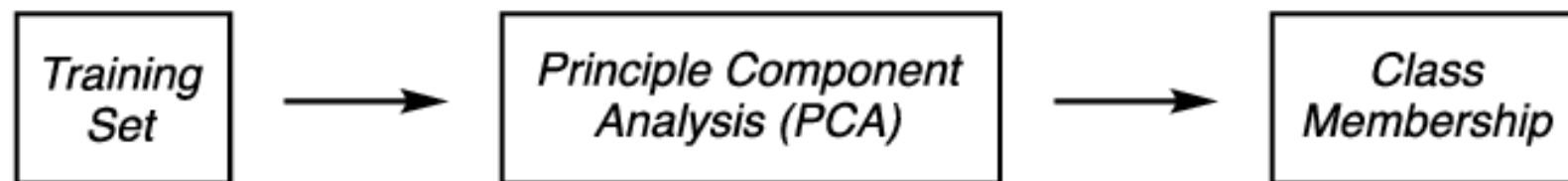
Soft Independent Modeling of Class Analogy, SIMCA

- ♦ SIMCA is a simple one-class classification method mainly based on PCA.
- ♦ The idea is create a PCA model only using sample belonging to a class and classify new samples based on how good the model can be fit them.
- ♦ The decision is made using the two distances: the orthogonal and score distances and corresponding critical limits.

Soft Independent Modelling of Class Analogy (SIMCA)



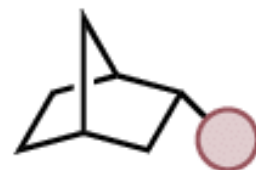
A SIMCA Workflow



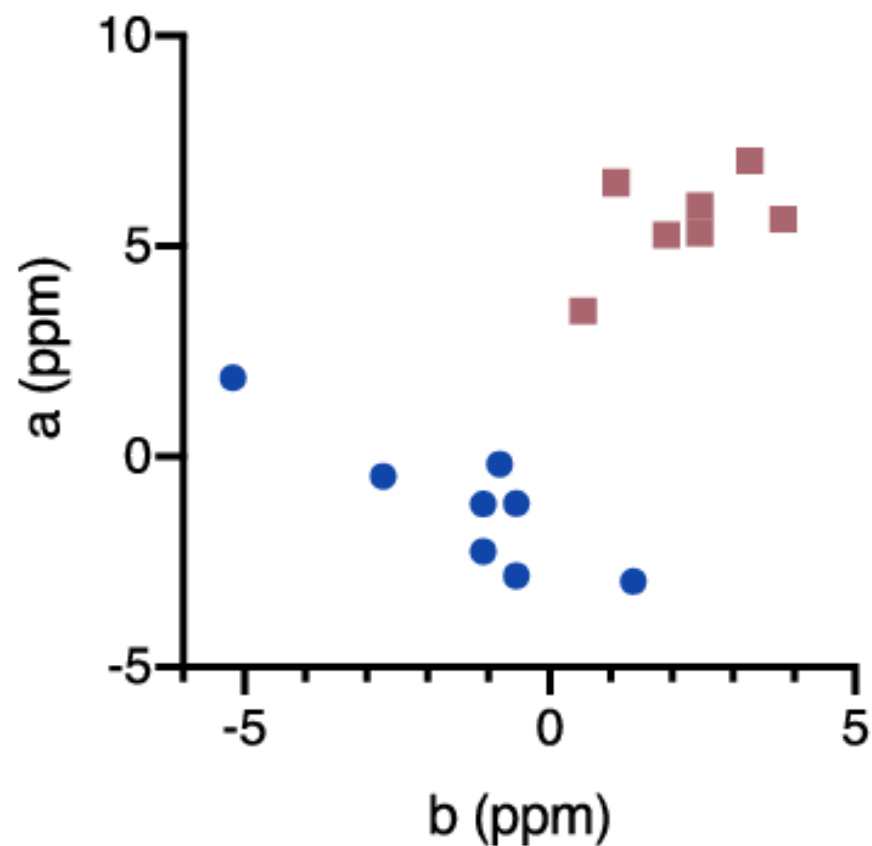
B Classification of Norboranes



exo



endo



Advantages of SIMCA

- ♦ SIMCA does not use a type of distribution assumption.
- ♦ SIMCA is a one-class method.
- ♦ The type I error can be adjusted.
- ♦ SIMCA is applicable to a large number of variables and resistant to the curse of dimensionality.

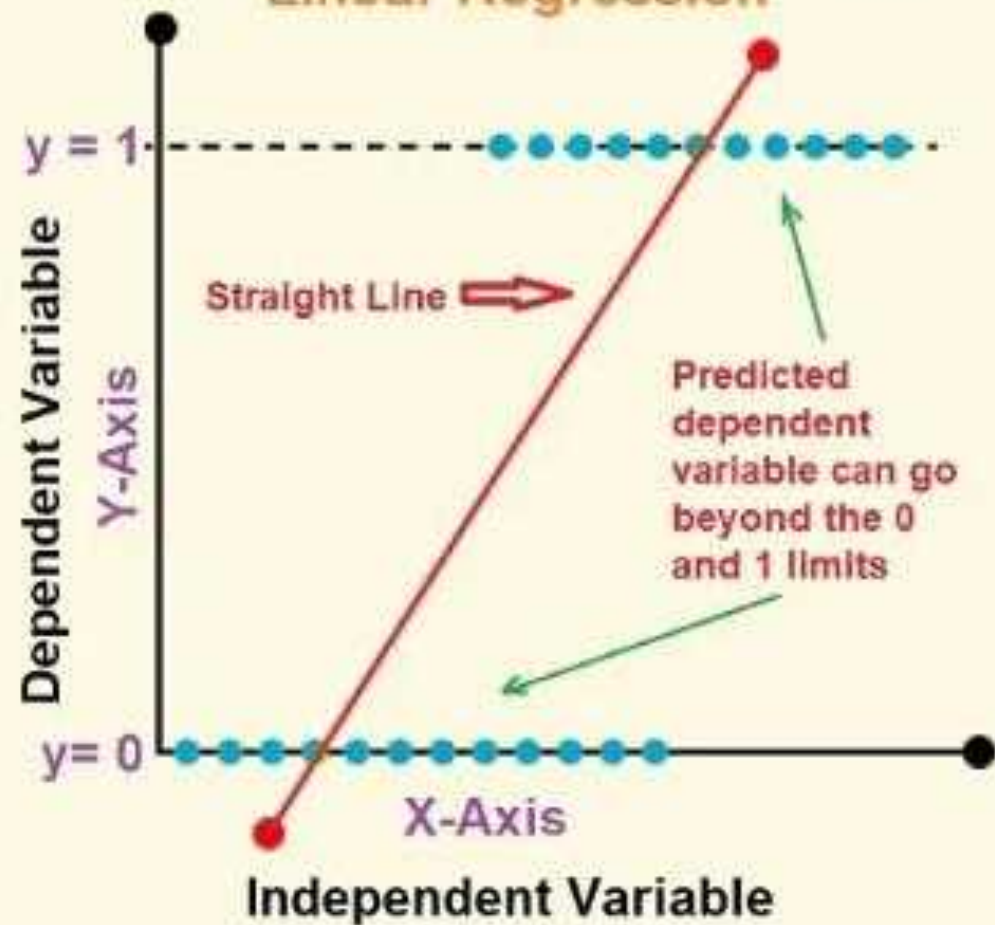
Disadvantages of SIMCA

- ◆ It requires a prior analysis by PCA.
- ◆ The results depend on the choice of PCs.
- ◆ However, this is facilitated by taking the minimum number for which the training set is correctly classified.
- ◆ SIMCA is sensitive to outliers, but the method can recognize them.
- ◆ SIMCA does not work for a small number of samples in the training set.

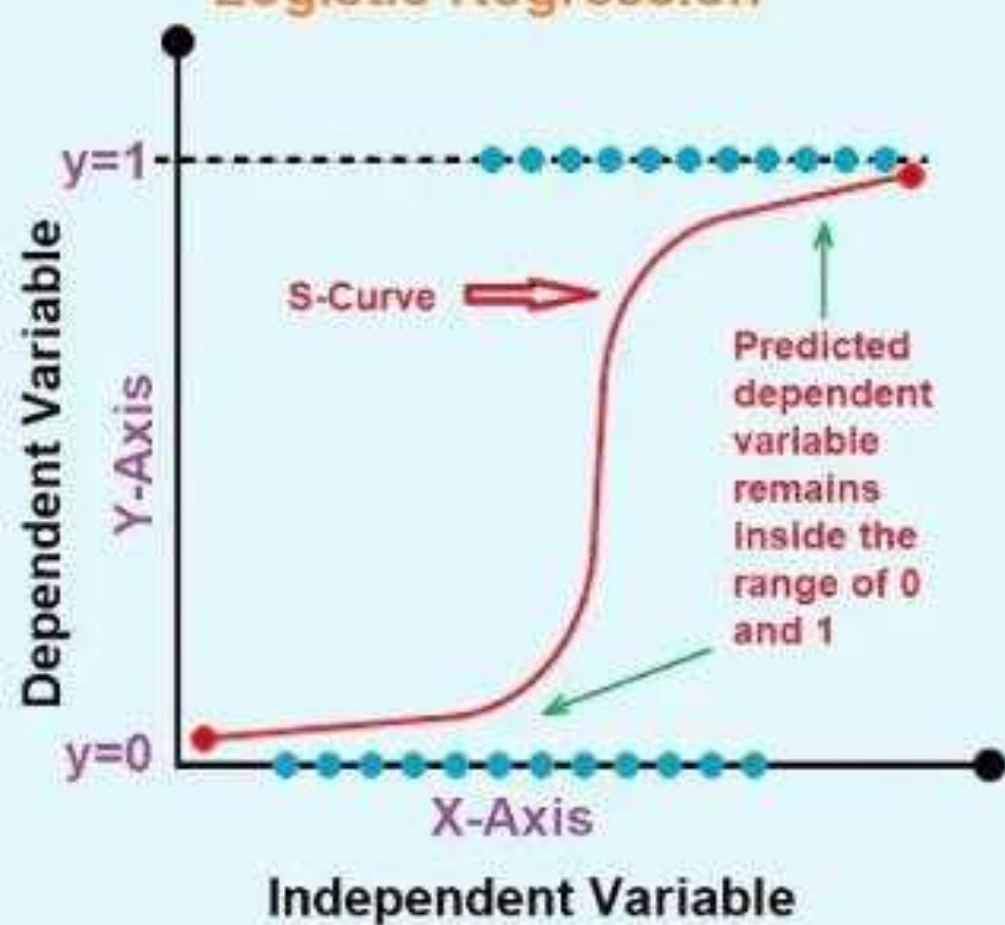
Logistic Regression

- ◆ Logistic Regression estimates the probability of an event occurring, such as voting or not voting, based on a given dataset of independent variables.

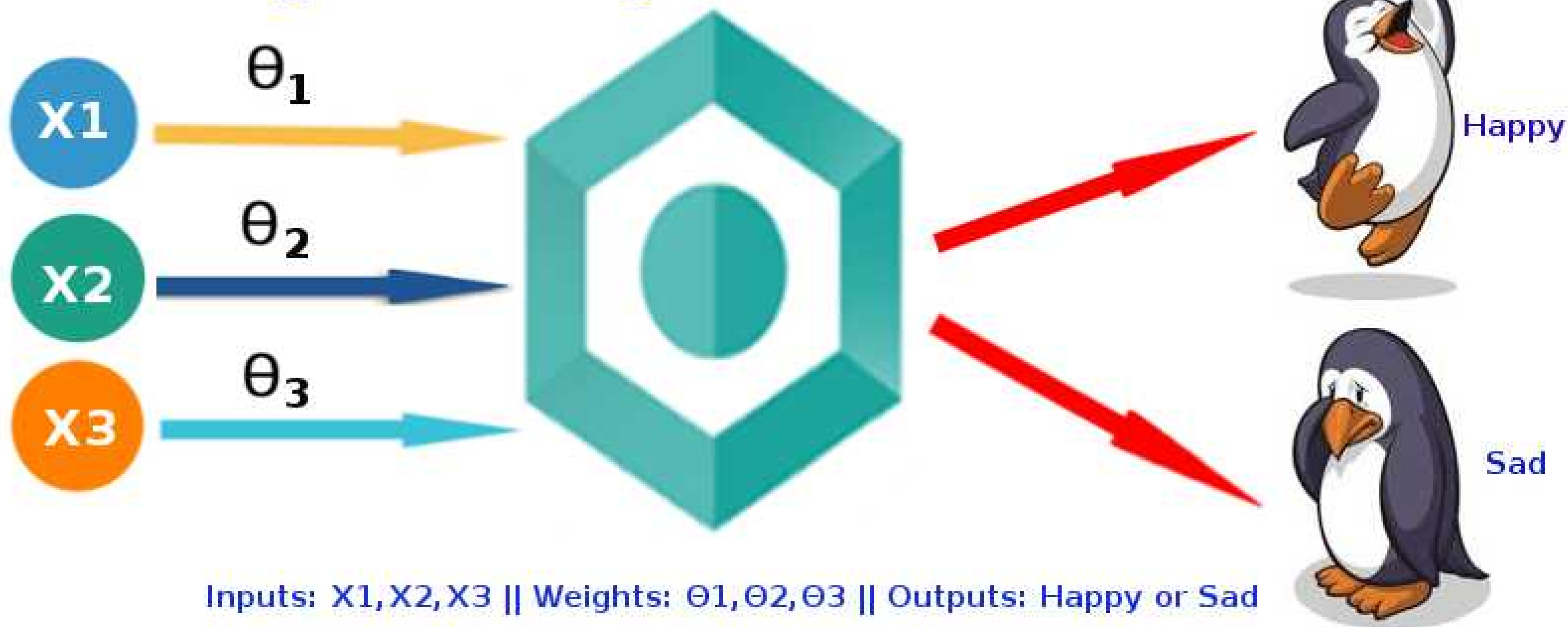
Linear Regression



Logistic Regression



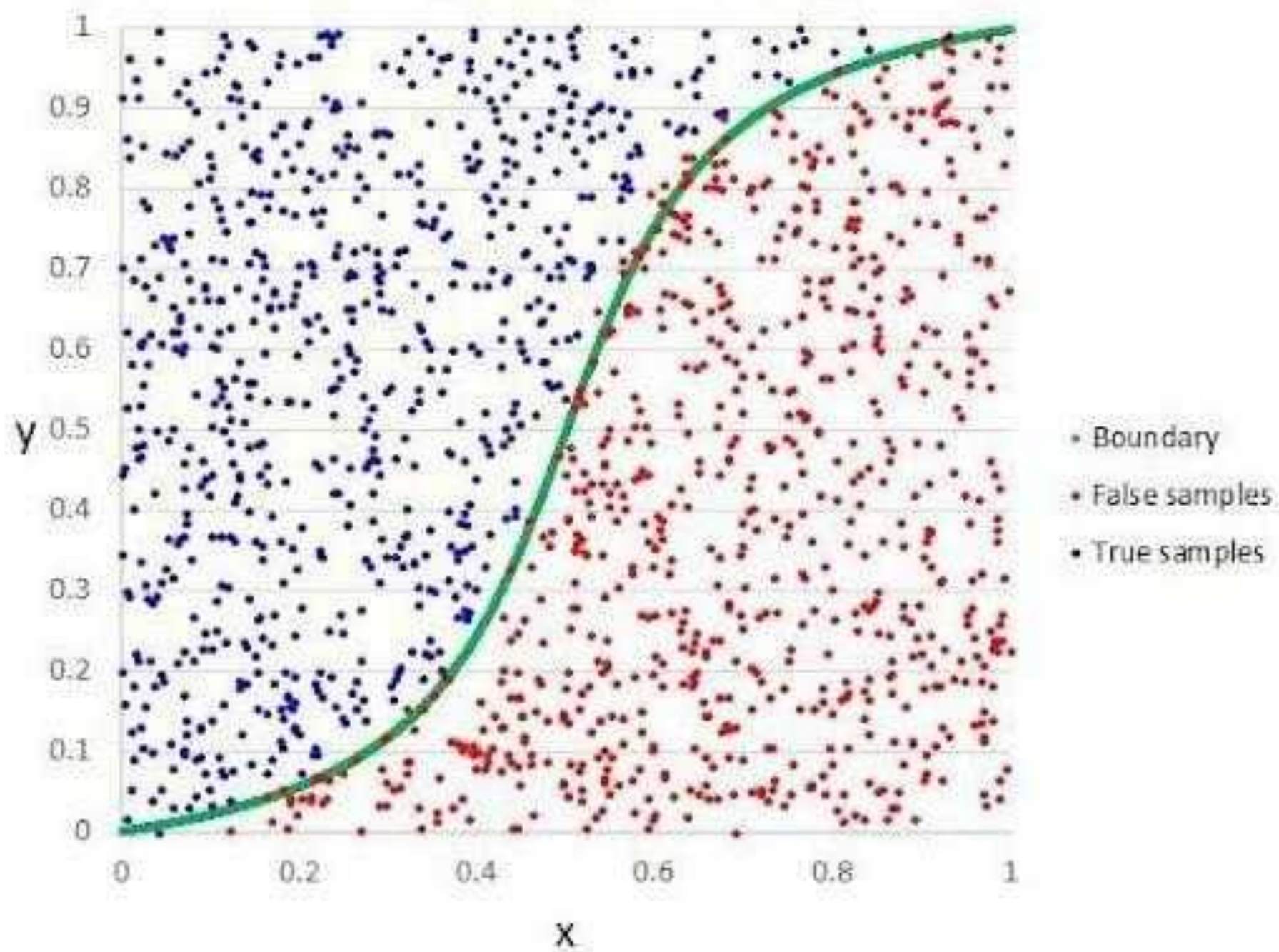
Logistic Regression Model



	Linear Regression	Logistic Regression
Response Variable	Continuous (e.g. price, age, height, distance)	Categorical (yes/no, male/female, win/not win)
Equation Used	$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots$	$p(Y) = e^{(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots)} / (1 + e^{(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots)})$
Method Used to Fit Equation	Ordinary Least Squares	Maximum Likelihood Estimation
Output to Predict	Continuous value (\$150, 40 years, 10 feet, etc.)	Probability (0.741, 0.122, 0.345, etc.)

$$P(Y = j) = \frac{\text{Exp}(\beta_j' x_i)}{1 + \sum_{k=1}^J \text{Exp}(\beta_k' x_i)} \text{ for } j = 1, \dots, J \text{ and}$$

	Linear Regression	Logistic Regression
Type of Variables Used	Numeric	Categorical
Curve Type	Straight Line	S type curve
Relationship between the dependent and independent variables	It requires a relationship between the dependent and independent variables	It doesn't require a relationship between the dependent and independent variables
Purpose	Linear regression is used to estimate the dependent variable in case of a change in independent variables.	Logistic regression is used to calculate the probability of an event
Distribution	Residuals are assumed to be normally distributed	Residuals need to be independent but not normally distributed
Examples	Relationship between number of hours worked and your salary.	Whether they are male or female



Types of logistic regression

- **BINARY LOGISTIC REGRESSION**

It is used when the dependent variable is dichotomous.

MULTINOMIAL LOGISTIC REGRESSION

— It is used when the dependent or outcomes variable has more than two categories.

LOGISTIC REGRESSION

Advantages and Disadvantages of Logistic Regression

Advantages



Logistic regression works well for cases where the dataset is linearly separable.



Logistic regression is much easier to implement than other methods, especially in the context of machine learning.



Logistic regression provides useful insights.

Disadvantages

The major drawback is, it is vulnerable to overfitting.



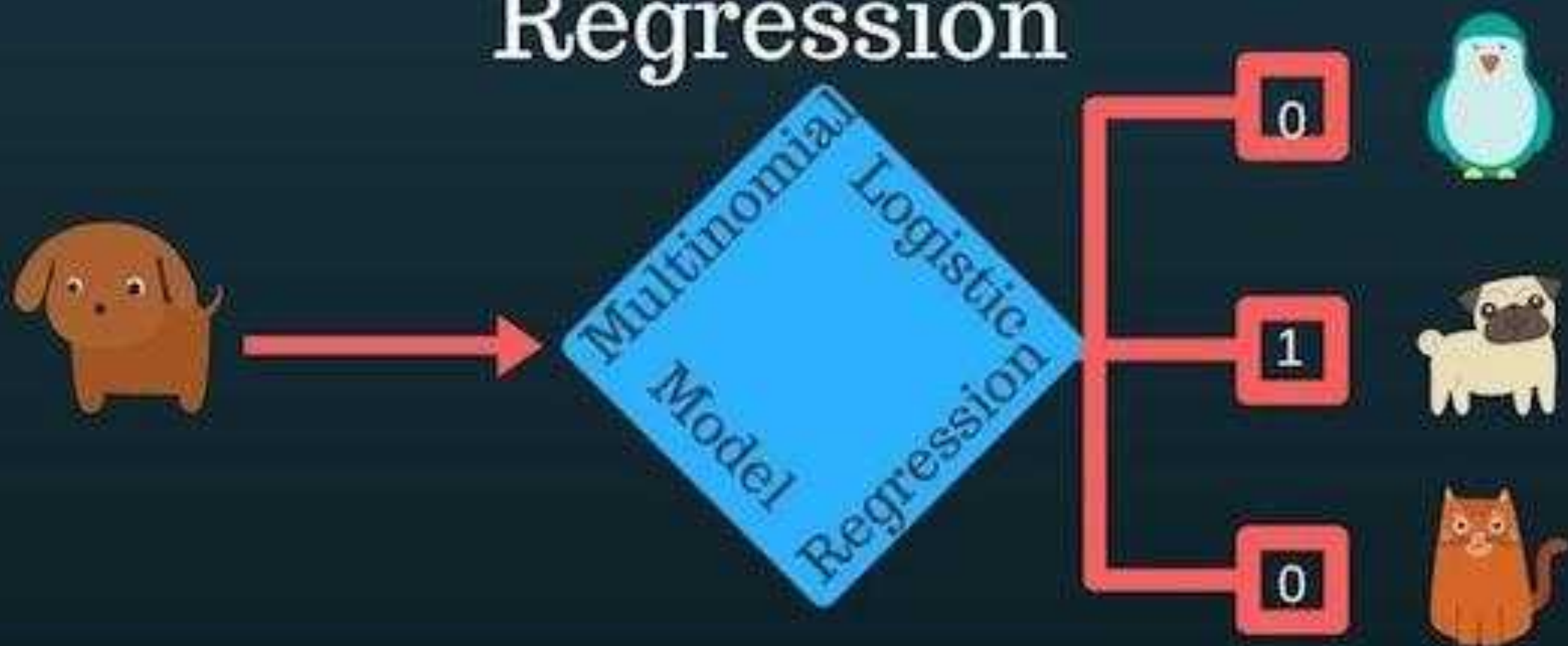
It can not handle many categorical variables, which means it cannot be used on a heavy model.



If independent variables are not correlated with the target variable, then this technique does not work properly.



Multinomial Logistic Regression



Bayes Classifier

- ◆ A probabilistic framework for solving classification problems
- ◆ Conditional Probability:

$$P(Y | X) = \frac{P(X, Y)}{P(X)}$$

$$P(X | Y) = \frac{P(X, Y)}{P(Y)}$$

- ◆ Bayes theorem:

$$P(Y | X) = \frac{P(X | Y)P(Y)}{P(X)}$$

Using Bayes Theorem for Classification

- ◆ Consider each attribute and class label as random variables
- ◆ Given a record with attributes $(X_1, X_2, \dots, X_d)$, the goal is to predict class Y

- Specifically, we want to find the value of Y that maximizes $P(Y | X_1, X_2, \dots, X_d)$

- ◆ Can we estimate $P(Y | X_1, X_2, \dots, X_d)$ directly from data?

<i>Tid</i>	Refund	Marital Status	Taxable Income	Evade
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Using Bayes Theorem for Classification

- ◆ Approach:

- compute posterior probability $P(Y | X_1, X_2, \dots, X_d)$ using the Bayes theorem

$$P(Y | X_1 X_2 \dots X_n) = \frac{P(X_1 X_2 \dots X_d | Y) P(Y)}{P(X_1 X_2 \dots X_d)}$$

- *Maximum a-posteriori*: Choose Y that maximizes
 $P(Y | X_1, X_2, \dots, X_d)$

- Equivalent to choosing value of Y that maximizes
 $P(X_1, X_2, \dots, X_d | Y) P(Y)$

- ◆ How to estimate $P(X_1, X_2, \dots, X_d | Y)$?

Example Data

Given a Test Record:

$X = (\text{Refund} = \text{No}, \text{Divorced}, \text{Income} = 120\text{K})$

<i>Tid</i>	Refund	Marital Status	Taxable Income	Evade
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

- We need to estimate $P(\text{Evade} = \text{Yes} \mid X)$ and $P(\text{Evade} = \text{No} \mid X)$

In the following we will replace
Evade = Yes by Yes, and
Evade = No by No

Example Data

Given a Test Record:

$X = (\text{Refund} = \text{No}, \text{Divorced}, \text{Income} = 120\text{K})$

<i>Tid</i>	Refund	Marital Status	Taxable Income	Evade
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Using Bayes Theorem:

$$P(\text{Yes}|X) = \frac{P(X|\text{Yes})P(\text{Yes})}{P(X)}$$

$$P(\text{No}|X) = \frac{P(X|\text{No})P(\text{No})}{P(X)}$$

How to estimate $P(X|\text{Yes})$ and $P(X|\text{No})$?

Conditional Independence

- ◆ X and Y are conditionally independent given Z if $P(X|YZ) = P(X|Z)$
- ◆ Example: Arm length and reading skills
 - Young child has shorter arm lengths and limited reading skills, compared to adults
 - If age is fixed, no apparent relationship between arm length and reading skills
 - Arm length and reading skills are conditionally independent given age

Naïve Bayes Classifier

- ◆ Assume independence among attributes X_i when class is given:
 - $P(X_1, X_2, \dots, X_d | Y_j) = P(X_1 | Y_j) P(X_2 | Y_j) \dots P(X_d | Y_j)$
 - Now we can estimate $P(X_i | Y_j)$ for all X_i and Y_j combinations from the training data
 - New point is classified to Y_j if $P(Y_j) \prod P(X_i | Y_j)$ is maximal.

Naïve Bayes on Example Data

Given a Test Record:

$X = (\text{Refund} = \text{No}, \text{Divorced}, \text{Income} = 120\text{K})$

<i>Tid</i>	Refund	Marital Status	Taxable Income	Evade
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

$P(X \mid \text{Yes}) =$

$P(\text{Refund} = \text{No} \mid \text{Yes}) \times$

$P(\text{Divorced} \mid \text{Yes}) \times$

$P(\text{Income} = 120\text{K} \mid \text{Yes})$

$P(X \mid \text{No}) =$

$P(\text{Refund} = \text{No} \mid \text{No}) \times$

$P(\text{Divorced} \mid \text{No}) \times$

$P(\text{Income} = 120\text{K} \mid \text{No})$

Estimate Probabilities from Data

<i>Tid</i>	Refund	Marital Status	Taxable Income	Evade
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

◇ $P(y)$ = fraction of instances of class y

- e.g., $P(\text{No}) = 7/10$,
 $P(\text{Yes}) = 3/10$

◇ For categorical attributes:

$$P(X_i = c | y) = n_c / n$$

- where $|X_i = c|$ is number of instances having attribute value $X_i = c$ and belonging to class y

- Examples:

$$P(\text{Status}=\text{Married}|\text{No}) = 4/7$$

$$P(\text{Refund}=\text{Yes}|\text{Yes})=0$$

Estimate Probabilities from Data

- ◆ For continuous attributes:
 - **Discretization:** Partition the range into bins:
 - ◆ Replace continuous value with bin value
 - ▢ Attribute changed from continuous to ordinal
 - **Probability density estimation:**
 - ◆ Assume attribute follows a normal distribution
 - ◆ Use data to estimate parameters of distribution (e.g., mean and standard deviation)
 - ◆ Once probability distribution is known, use it to estimate the conditional probability $P(X_i|Y)$

Estimate Probabilities from Data

<i>Tid</i>	Refund	Marital Status	Taxable Income	Evade
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

◆ Normal distribution:

$$P(X_i | Y_j) = \frac{1}{\sqrt{2\pi\sigma_{ij}^2}} e^{-\frac{(X_i - \mu_{ij})^2}{2\sigma_{ij}^2}}$$

- One for each (X_i, Y_i) pair

◆ For (Income, Class=No):

- If Class=No

◆ sample mean = 110

◆ sample variance = 2975

$$P(\text{Income} = 120 | \text{No}) = \frac{1}{\sqrt{2\pi(54.54)}} e^{-\frac{(120-110)^2}{2(2975)}} = 0.0072$$

Example of Naïve Bayes Classifier

Given a Test Record:

$X = (\text{Refund} = \text{No}, \text{Divorced}, \text{Income} = 120\text{K})$

Naïve Bayes Classifier:

$$P(\text{Refund} = \text{Yes} \mid \text{No}) = 3/7$$

$$P(\text{Refund} = \text{No} \mid \text{No}) = 4/7$$

$$P(\text{Refund} = \text{Yes} \mid \text{Yes}) = 0$$

$$P(\text{Refund} = \text{No} \mid \text{Yes}) = 1$$

$$P(\text{Marital Status} = \text{Single} \mid \text{No}) = 2/7$$

$$P(\text{Marital Status} = \text{Divorced} \mid \text{No}) = 1/7$$

$$P(\text{Marital Status} = \text{Married} \mid \text{No}) = 4/7$$

$$P(\text{Marital Status} = \text{Single} \mid \text{Yes}) = 2/3$$

$$P(\text{Marital Status} = \text{Divorced} \mid \text{Yes}) = 1/3$$

$$P(\text{Marital Status} = \text{Married} \mid \text{Yes}) = 0$$

For Taxable Income:

If class = No: sample mean = 110

sample variance = 2975

If class = Yes: sample mean = 90

sample variance = 25

- $P(X \mid \text{No}) = P(\text{Refund} = \text{No} \mid \text{No})$
 $\times P(\text{Divorced} \mid \text{No})$
 $\times P(\text{Income} = 120\text{K} \mid \text{No})$
 $= 4/7 \times 1/7 \times 0.0072 = 0.0006$
- $P(X \mid \text{Yes}) = P(\text{Refund} = \text{No} \mid \text{Yes})$
 $\times P(\text{Divorced} \mid \text{Yes})$
 $\times P(\text{Income} = 120\text{K} \mid \text{Yes})$
 $= 1 \times 1/3 \times 1.2 \times 10^{-9} = 4 \times 10^{-10}$

Since $P(X \mid \text{No})P(\text{No}) > P(X \mid \text{Yes})P(\text{Yes})$

Therefore $P(\text{No} \mid X) > P(\text{Yes} \mid X)$

$\Rightarrow \text{Class} = \text{No}$

Naïve Bayes Classifier can make decisions with partial information about attributes in the test record

Even in absence of information about any attributes, we can use Apriori Probabilities of Class Variable:

Naïve Bayes Classifier:

$$P(\text{Refund} = \text{Yes} \mid \text{No}) = 3/7$$

$$P(\text{Refund} = \text{No} \mid \text{No}) = 4/7$$

$$P(\text{Refund} = \text{Yes} \mid \text{Yes}) = 0$$

$$P(\text{Refund} = \text{No} \mid \text{Yes}) = 1$$

$$P(\text{Marital Status} = \text{Single} \mid \text{No}) = 2/7$$

$$P(\text{Marital Status} = \text{Divorced} \mid \text{No}) = 1/7$$

$$P(\text{Marital Status} = \text{Married} \mid \text{No}) = 4/7$$

$$P(\text{Marital Status} = \text{Single} \mid \text{Yes}) = 2/3$$

$$P(\text{Marital Status} = \text{Divorced} \mid \text{Yes}) = 1/3$$

$$P(\text{Marital Status} = \text{Married} \mid \text{Yes}) = 0$$

For Taxable Income:

If class = No: sample mean = 110

sample variance = 2975

If class = Yes: sample mean = 90

sample variance = 25

$$P(\text{Yes}) = 3/10$$

$$P(\text{No}) = 7/10$$

If we only know that marital status is Divorced, then:

$$P(\text{Yes} \mid \text{Divorced}) = 1/3 \times 3/10 / P(\text{Divorced})$$

$$P(\text{No} \mid \text{Divorced}) = 1/7 \times 7/10 / P(\text{Divorced})$$

If we also know that Refund = No, then

$$P(\text{Yes} \mid \text{Refund} = \text{No}, \text{Divorced}) = 1 \times 1/3 \times 3/10 / P(\text{Divorced}, \text{Refund} = \text{No})$$

$$P(\text{No} \mid \text{Refund} = \text{No}, \text{Divorced}) = 4/7 \times 1/7 \times 7/10 / P(\text{Divorced}, \text{Refund} = \text{No})$$

If we also know that Taxable Income = 120, then

$$P(\text{Yes} \mid \text{Refund} = \text{No}, \text{Divorced}, \text{Income} = 120) = 1.2 \times 10^{-9} \times 1 \times 1/3 \times 3/10 / P(\text{Divorced}, \text{Refund} = \text{No}, \text{Income} = 120)$$

$$P(\text{No} \mid \text{Refund} = \text{No}, \text{Divorced}, \text{Income} = 120) = 0.0072 \times 4/7 \times 1/7 \times 7/10 / P(\text{Divorced}, \text{Refund} = \text{No}, \text{Income} = 120)$$

Issues with Naïve Bayes Classifier

Given a Test Record: $X = (\text{Married})$

Naïve Bayes Classifier:

$$P(\text{Refund} = \text{Yes} \mid \text{No}) = 3/7$$

$$P(\text{Refund} = \text{No} \mid \text{No}) = 4/7$$

$$P(\text{Refund} = \text{Yes} \mid \text{Yes}) = 0$$

$$P(\text{Refund} = \text{No} \mid \text{Yes}) = 1$$

$$P(\text{Marital Status} = \text{Single} \mid \text{No}) = 2/7$$

$$P(\text{Marital Status} = \text{Divorced} \mid \text{No}) = 1/7$$

$$P(\text{Marital Status} = \text{Married} \mid \text{No}) = 4/7$$

$$P(\text{Marital Status} = \text{Single} \mid \text{Yes}) = 2/3$$

$$P(\text{Marital Status} = \text{Divorced} \mid \text{Yes}) = 1/3$$

$$P(\text{Marital Status} = \text{Married} \mid \text{Yes}) = 0$$

For Taxable Income:

If class = No: sample mean = 110

sample variance = 2975

If class = Yes: sample mean = 90

sample variance = 25

$$P(\text{Yes}) = 3/10$$

$$P(\text{No}) = 7/10$$

$$P(\text{Yes} \mid \text{Married}) = 0 \times 3/10 / P(\text{Married})$$

$$P(\text{No} \mid \text{Married}) = 4/7 \times 7/10 / P(\text{Married})$$

Issues with Naïve Bayes Classifier

Consider the table with Tid = 7 deleted

Tid	Refund	Marital Status	Taxable Income	Evade
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Naïve Bayes Classifier:

$$P(\text{Refund} = \text{Yes} \mid \text{No}) = 2/6$$

$$P(\text{Refund} = \text{No} \mid \text{No}) = 4/6$$

$$P(\text{Refund} = \text{Yes} \mid \text{Yes}) = 0$$

$$P(\text{Refund} = \text{No} \mid \text{Yes}) = 1$$

$$P(\text{Marital Status} = \text{Single} \mid \text{No}) = 2/6$$

$$P(\text{Marital Status} = \text{Divorced} \mid \text{No}) = 0$$

$$P(\text{Marital Status} = \text{Married} \mid \text{No}) = 4/6$$

$$P(\text{Marital Status} = \text{Single} \mid \text{Yes}) = 2/3$$

$$P(\text{Marital Status} = \text{Divorced} \mid \text{Yes}) = 1/3$$

$$P(\text{Marital Status} = \text{Married} \mid \text{Yes}) = 0/3$$

For Taxable Income:

If class = No: sample mean = 91

sample variance = 685

If class = No: sample mean = 90

sample variance = 25

Given $X = (\text{Refund} = \text{Yes}, \text{Divorced}, 120\text{K})$

$$P(X \mid \text{No}) = 2/6 \times 0 \times 0.0083 = 0$$

$$P(X \mid \text{Yes}) = 0 \times 1/3 \times 1.2 \times 10^{-9} = 0$$

Naïve Bayes will not be able to
classify X as Yes or No!

Issues with Naïve Bayes Classifier

- ◆ If one of the conditional probabilities is zero, then the entire expression becomes zero
- ◆ Need to use other estimates of conditional probabilities than simple fractions
- ◆ Probability estimation:

$$\text{original: } P(X_i = c|y) = \frac{n_c}{n}$$

$$\text{Laplace Estimate: } P(X_i = c|y) = \frac{n_c + 1}{n + v}$$

$$\text{m - estimate: } P(X_i = c|y) = \frac{n_c + mp}{n + m}$$

n : number of training instances belonging to class y

n_c : number of instances with $X_i = c$ and $Y = y$

v : total number of attribute values that X_i can take

p : initial estimate of
($P(X_i = c|y)$ known apriori)

m : hyper-parameter for our confidence in p

Example of Naïve Bayes Classifier

Name	Give Birth	Can Fly	Live in Water	Have Legs	Class
human	yes	no	no	yes	mammals
python	no	no	no	no	non-mammals
salmon	no	no	yes	no	non-mammals
whale	yes	no	yes	no	mammals
frog	no	no	sometimes	yes	non-mammals
komodo	no	no	no	yes	non-mammals
bat	yes	yes	no	yes	mammals
pigeon	no	yes	no	yes	non-mammals
cat	yes	no	no	yes	mammals
leopard shark	yes	no	yes	no	non-mammals
turtle	no	no	sometimes	yes	non-mammals
penguin	no	no	sometimes	yes	non-mammals
porcupine	yes	no	no	yes	mammals
eel	no	no	yes	no	non-mammals
salamander	no	no	sometimes	yes	non-mammals
gila monster	no	no	no	yes	non-mammals
platypus	no	no	no	yes	mammals
owl	no	yes	no	yes	non-mammals
dolphin	yes	no	yes	no	mammals
eagle	no	yes	no	yes	non-mammals

A: attributes

M: mammals

N: non-mammals

$$P(A | M) = \frac{6}{7} \times \frac{6}{7} \times \frac{2}{7} \times \frac{2}{7} = 0.06$$

$$P(A | N) = \frac{1}{13} \times \frac{10}{13} \times \frac{3}{13} \times \frac{4}{13} = 0.0042$$

$$P(A | M)P(M) = 0.06 \times \frac{7}{20} = 0.021$$

$$P(A | N)P(N) = 0.004 \times \frac{13}{20} = 0.0027$$

Give Birth	Can Fly	Live in Water	Have Legs	Class
yes	no	yes	no	?

P(A|M)P(M) > P(A|N)P(N)

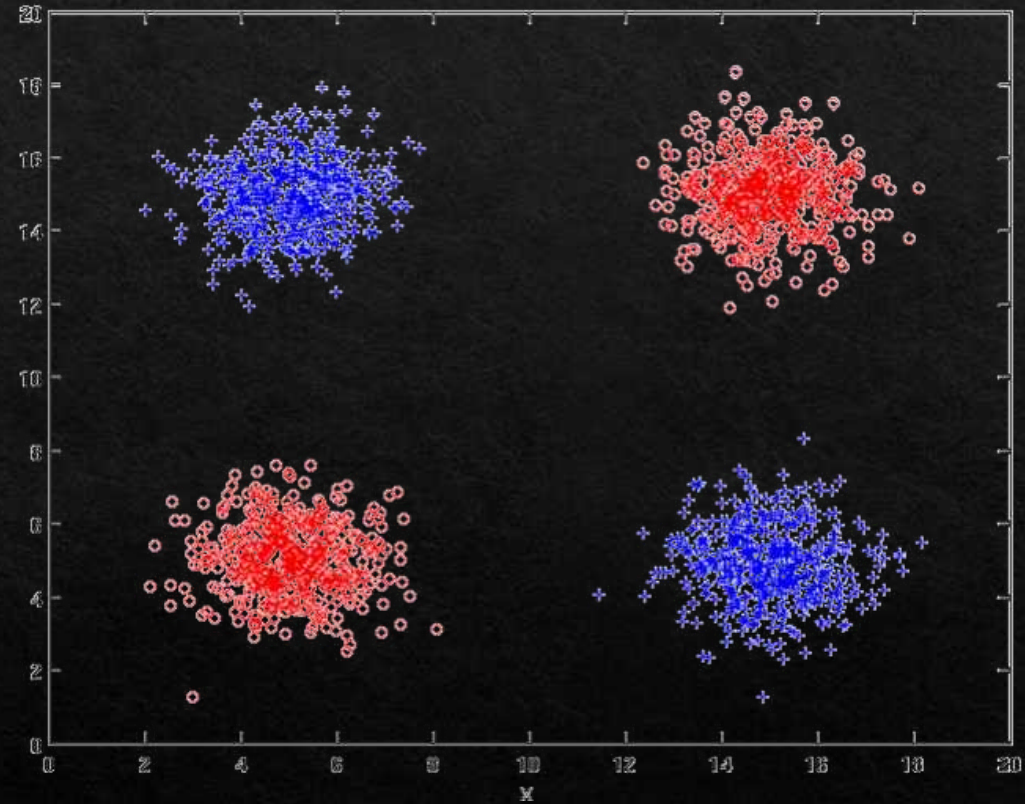
=> Mammals

Naïve Bayes (Summary)

- ◆ Robust to isolated noise points
- ◆ Handle missing values by ignoring the instance during probability estimate calculations
- ◆ Robust to irrelevant attributes
- ◆ Redundant and correlated attributes will violate class conditional assumption
 - Use other techniques such as Bayesian Belief Networks (BBN)

Naïve Bayes

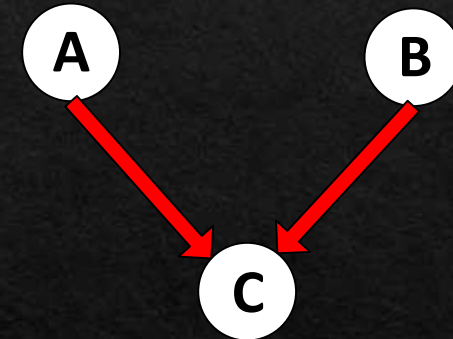
- ◆ How does Naïve Bayes perform on the following dataset?



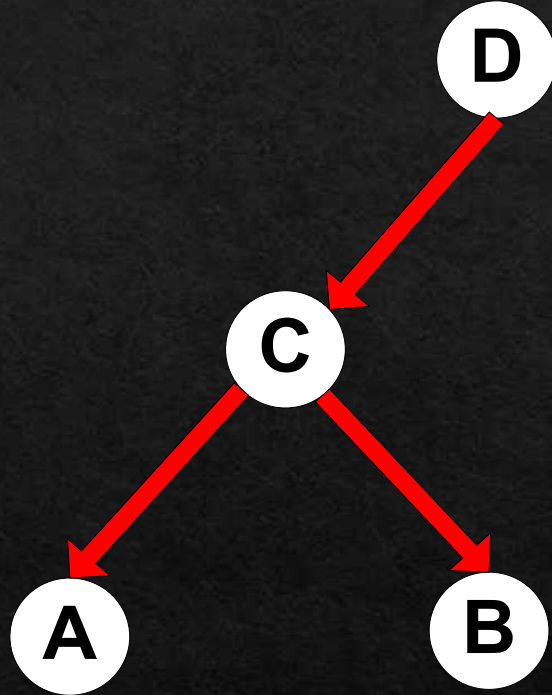
Conditional independence of attributes is violated

Bayesian Belief Networks

- ◆ Provides graphical representation of probabilistic relationships among a set of random variables
- ◆ Consists of:
 - A directed acyclic graph (dag)
 - ◆ Node corresponds to a variable
 - ◆ Arc corresponds to dependence relationship between a pair of variables
 - A probability table associating each node to its immediate parent



Conditional Independence



D is parent of C

A is child of C

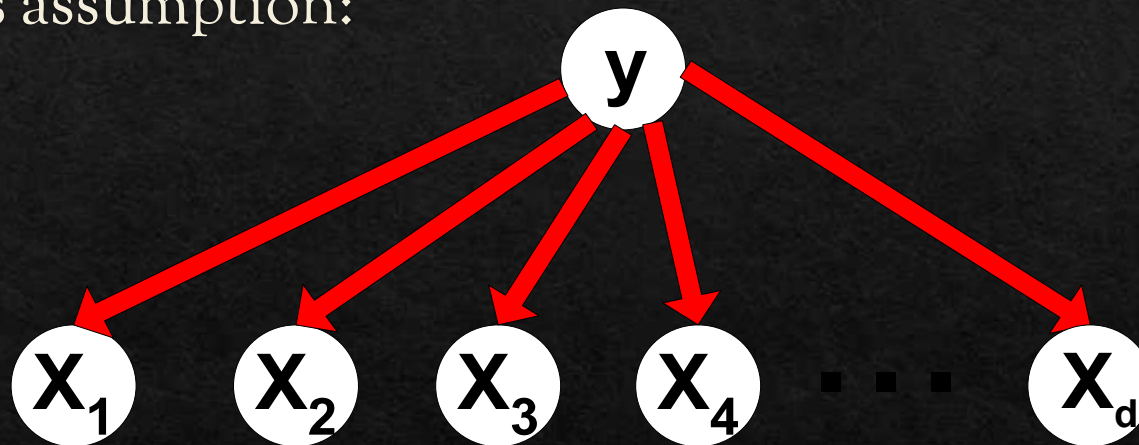
B is descendant of D

D is ancestor of A

- ◆ A node in a Bayesian network is conditionally independent of all of its nondescendants, if its parents are known

Conditional Independence

◊ Naïve Bayes assumption:

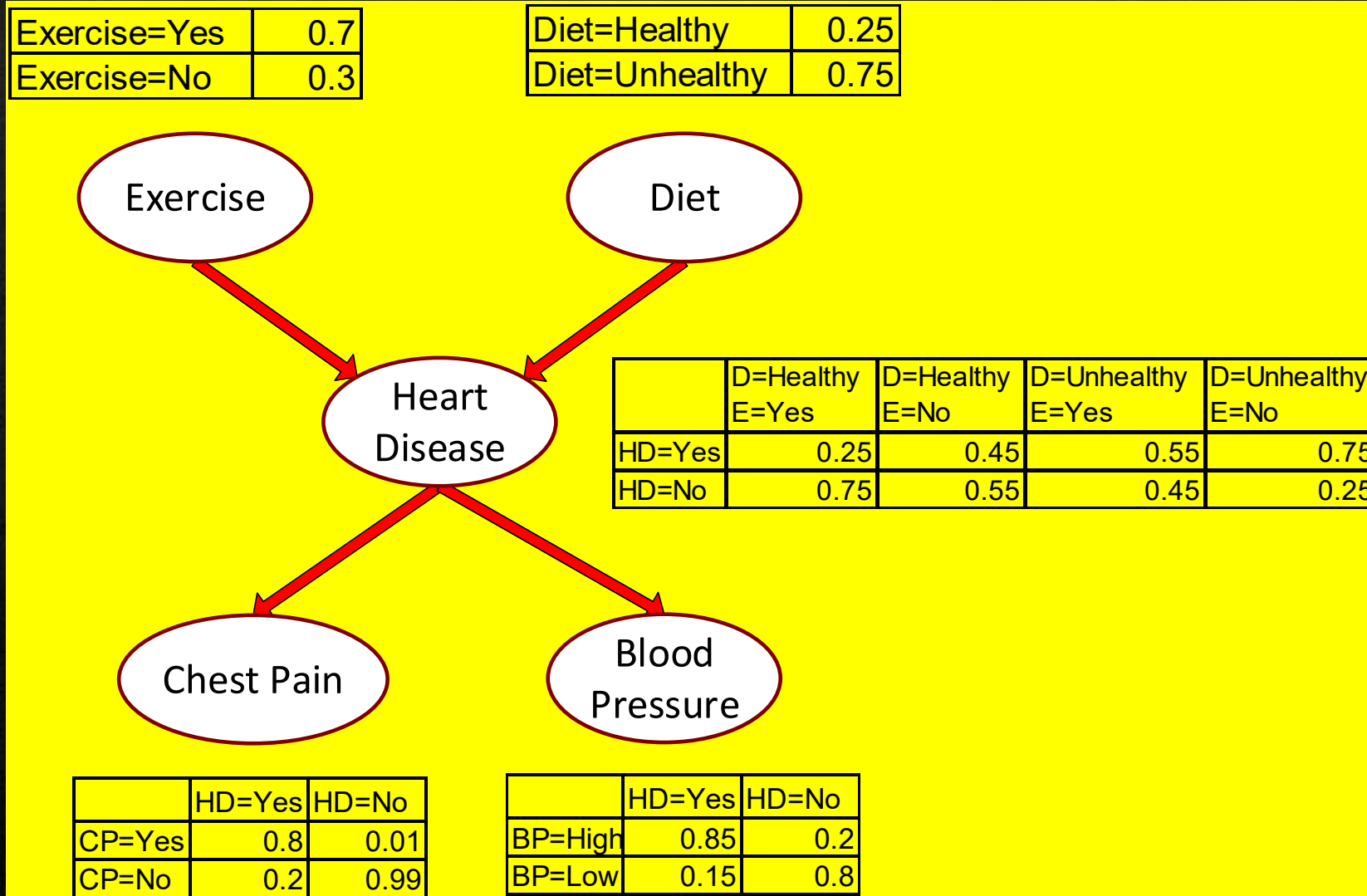


Probability Tables

- ◊ If X does not have any parents, table contains prior probability $P(X)$
- ◊ If X has only one parent (Y), table contains conditional probability $P(X|Y)$
- ◊ If X has multiple parents ($Y_1, Y_2, \dots, Y_k$), table contains conditional probability $P(X|Y_1, Y_2, \dots, Y_k)$



Example of Bayesian Belief Network



Example of Inferencing using BBN

- ◆ Given: $X = (E=No, D=Yes, CP=Yes, BP=High)$
 - Compute $P(HD|E,D,CP,BP)$?
- ◆ $P(HD=Yes| E=No,D=Yes) = 0.55$
 $P(CP=Yes| HD=Yes) = 0.8$
 $P(BP=High| HD=Yes) = 0.85$
 - $P(HD=Yes|E=No,D=Yes,CP=Yes,BP=High)$
 $\propto 0.55 \times 0.8 \times 0.85 = 0.374$
- ◆ $P(HD=No| E=No,D=Yes) = 0.45$
 $P(CP=Yes| HD=No) = 0.01$
 $P(BP=High| HD=No) = 0.2$
 - $P(HD=No|E=No,D=Yes,CP=Yes,BP=High)$
 $\propto 0.45 \times 0.01 \times 0.2 = 0.0009$



**Classify X
as Yes**

Rule-Based Classifier

- ◆ Classify records by using a collection of “if...then...” rules
- ◆ Rule: $(Condition) \rightarrow y$
 - where
 - ◆ *Condition* is a conjunction of tests on attributes
 - ◆ y is the class label
 - Examples of classification rules:
 - ◆ $(\text{Blood Type}=\text{Warm}) \wedge (\text{Lay Eggs}=\text{Yes}) \rightarrow \text{Birds}$
 - ◆ $(\text{Taxable Income} < 50K) \wedge (\text{Refund}=\text{Yes}) \rightarrow \text{Evade}=\text{No}$

Rule-based Classifier (Example)

Name	Blood Type	Give Birth	Can Fly	Live in Water	Class
human	warm	yes	no	no	mammals
python	cold	no	no	no	reptiles
salmon	cold	no	no	yes	fishes
whale	warm	yes	no	yes	mammals
frog	cold	no	no	sometimes	amphibians
komodo	cold	no	no	no	reptiles
bat	warm	yes	yes	no	mammals
pigeon	warm	no	yes	no	birds
cat	warm	yes	no	no	mammals
leopard shark	cold	yes	no	yes	fishes
turtle	cold	no	no	sometimes	reptiles
penguin	warm	no	no	sometimes	birds
porcupine	warm	yes	no	no	mammals
eel	cold	no	no	yes	fishes
salamander	cold	no	no	sometimes	amphibians
gila monster	cold	no	no	no	reptiles
platypus	warm	no	no	no	mammals
owl	warm	no	yes	no	birds
dolphin	warm	yes	no	yes	mammals
eagle	warm	no	yes	no	birds

R₁: (Give Birth = no) $\wedge$ (Can Fly = yes) $\rightarrow$ Birds

R₂: (Give Birth = no) $\wedge$ (Live in Water = yes) $\rightarrow$ Fishes

R₃: (Give Birth = yes) $\wedge$ (Blood Type = warm) $\rightarrow$ Mammals

R₄: (Give Birth = no) $\wedge$ (Can Fly = no) $\rightarrow$ Reptiles

R₅: (Live in Water = sometimes) $\rightarrow$ Amphibians

Application of Rule-Based Classifier

- ♦ A rule r **covers** an instance x if the attributes of the instance satisfy the condition of the rule

R1: (Give Birth = no) $\wedge$ (Can Fly = yes) $\rightarrow$ Birds

R2: (Give Birth = no) $\wedge$ (Live in Water = yes) $\rightarrow$ Fishes

R3: (Give Birth = yes) $\wedge$ (Blood Type = warm) $\rightarrow$ Mammals

R4: (Give Birth = no) $\wedge$ (Can Fly = no) $\rightarrow$ Reptiles

R5: (Live in Water = sometimes) $\rightarrow$ Amphibians

Name	Blood Type	Give Birth	Can Fly	Live in Water	Class
hawk	warm	no	yes	no	?
grizzly bear	warm	yes	no	no	?

The rule R1 covers a hawk $\Rightarrow$ Bird

The rule R3 covers the grizzly bear $\Rightarrow$ Mammal

Rule Coverage and Accuracy

- ◆ Coverage of a rule:
 - Fraction of records that satisfy the antecedent of a rule
- ◆ Accuracy of a rule:
 - Fraction of records that satisfy the antecedent that also satisfy the consequent of a rule

<i>Tid</i>	Refund	Marital Status	Taxable Income	Class
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

(Status=Single) → No

Coverage = 40%, Accuracy = 50%

How does Rule-based Classifier Work?

R1: (Give Birth = no) $\wedge$ (Can Fly = yes) $\rightarrow$ Birds

R2: (Give Birth = no) $\wedge$ (Live in Water = yes) $\rightarrow$ Fishes

R3: (Give Birth = yes) $\wedge$ (Blood Type = warm) $\rightarrow$ Mammals

R4: (Give Birth = no) $\wedge$ (Can Fly = no) $\rightarrow$ Reptiles

R5: (Live in Water = sometimes) $\rightarrow$ Amphibians

Name	Blood Type	Give Birth	Can Fly	Live in Water	Class
lemur	warm	yes	no	no	?
turtle	cold	no	no	sometimes	?
dogfish shark	cold	yes	no	yes	?

A lemur triggers rule R3, so it is classified as a mammal

A turtle triggers both R4 and R5

A dogfish shark triggers none of the rules

Characteristics of Rule Sets: Strategy 1

- ◆ Mutually exclusive rules
 - Classifier contains mutually exclusive rules if the rules are independent of each other
 - Every record is covered by at most one rule
- ◆ Exhaustive rules
 - Classifier has exhaustive coverage if it accounts for every possible combination of attribute values
 - Each record is covered by at least one rule

Characteristics of Rule Sets: Strategy 2

- ◆ Rules are not mutually exclusive
 - A record may trigger more than one rule
 - Solution?
 - ◆ Ordered rule set
 - ◆ Unordered rule set – use voting schemes
- ◆ Rules are not exhaustive
 - A record may not trigger any rules
 - Solution?
 - ◆ Use a default class

Ordered Rule Set

- ◆ Rules are rank ordered according to their priority
 - An ordered rule set is known as a decision list
- ◆ When a test record is presented to the classifier
 - It is assigned to the class label of the highest ranked rule it has triggered
 - If none of the rules fired, it is assigned to the default class

R1: (Give Birth = no) $\wedge$ (Can Fly = yes) $\rightarrow$ Birds

R2: (Give Birth = no) $\wedge$ (Live in Water = yes) $\rightarrow$ Fishes

R3: (Give Birth = yes) $\wedge$ (Blood Type = warm) $\rightarrow$ Mammals

R4: (Give Birth = no) $\wedge$ (Can Fly = no) $\rightarrow$ Reptiles

R5: (Live in Water = sometimes) $\rightarrow$ Amphibians



Name	Blood Type	Give Birth	Can Fly	Live in Water	Class
turtle	cold	no	no	sometimes	?

Rule Ordering Schemes

- ◆ Rule-based ordering
 - Individual rules are ranked based on their quality
- ◆ Class-based ordering
 - Rules that belong to the same class appear together

Rule-based Ordering

(Refund=Yes) ==> No

(Refund=No, Marital Status={Single,Divorced},
Taxable Income<80K) ==> No

(Refund=No, Marital Status={Single,Divorced},
Taxable Income>80K) ==> Yes

(Refund=No, Marital Status={Married}) ==> No

Class-based Ordering

(Refund=Yes) ==> No

(Refund=No, Marital Status={Single,Divorced},
Taxable Income<80K) ==> No

(Refund=No, Marital Status={Married}) ==> No

(Refund=No, Marital Status={Single,Divorced},
Taxable Income>80K) ==> Yes

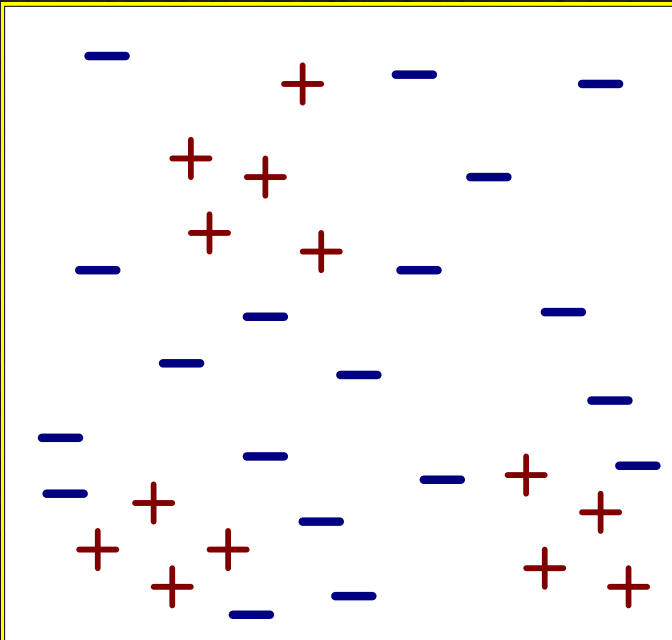
Building Classification Rules

- ◆ Direct Method:
 - ◆ Extract rules directly from data
 - ◆ Examples: RIPPER, CN₂, Holte's 1R
- ◆ Indirect Method:
 - ◆ Extract rules from other classification models (e.g. decision trees, neural networks, etc).
 - ◆ Examples: C_{4.5}rules

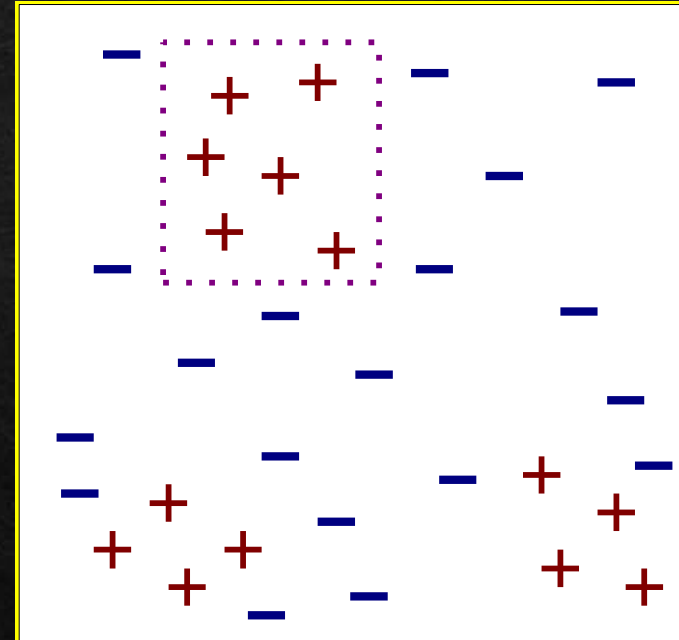
Direct Method: Sequential Covering

1. Start from an empty rule
2. Grow a rule using the Learn-One-Rule function
3. Remove training records covered by the rule
4. Repeat Step (2) and (3) until stopping criterion is met

Example of Sequential Covering

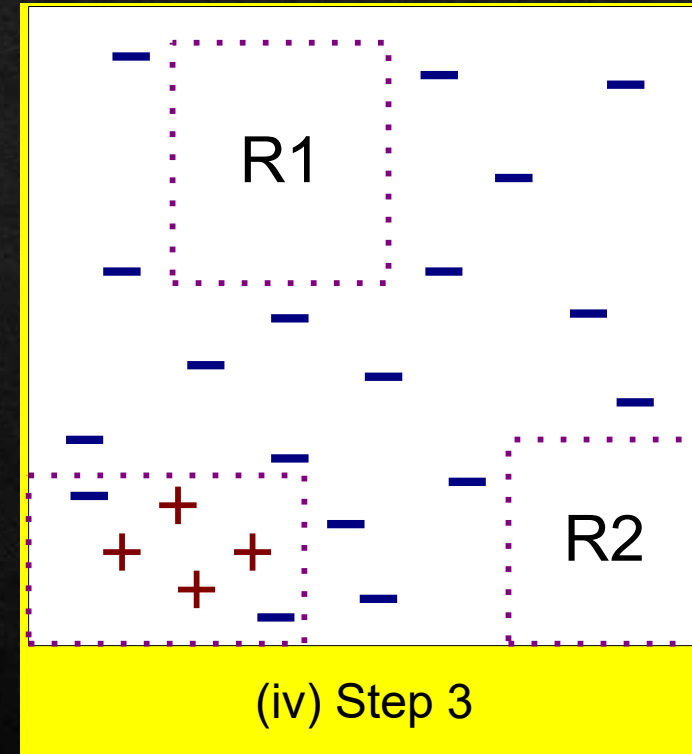
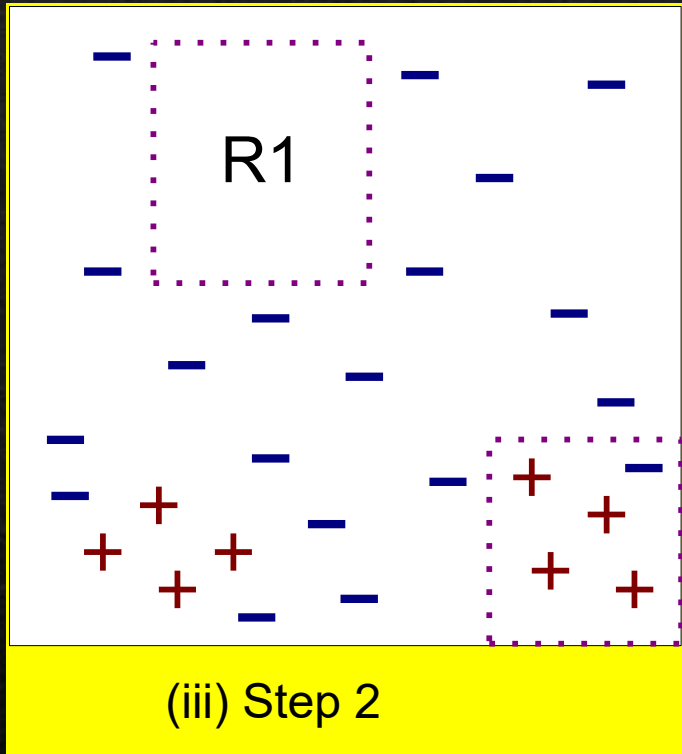


(i) Original Data



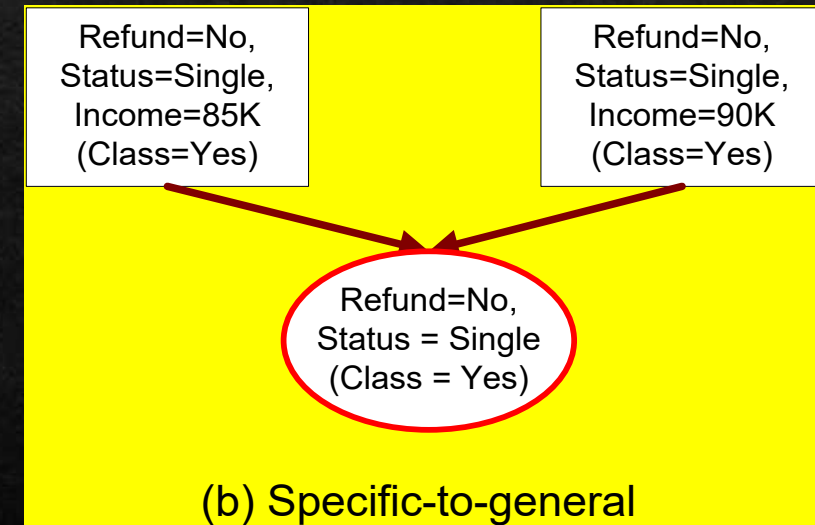
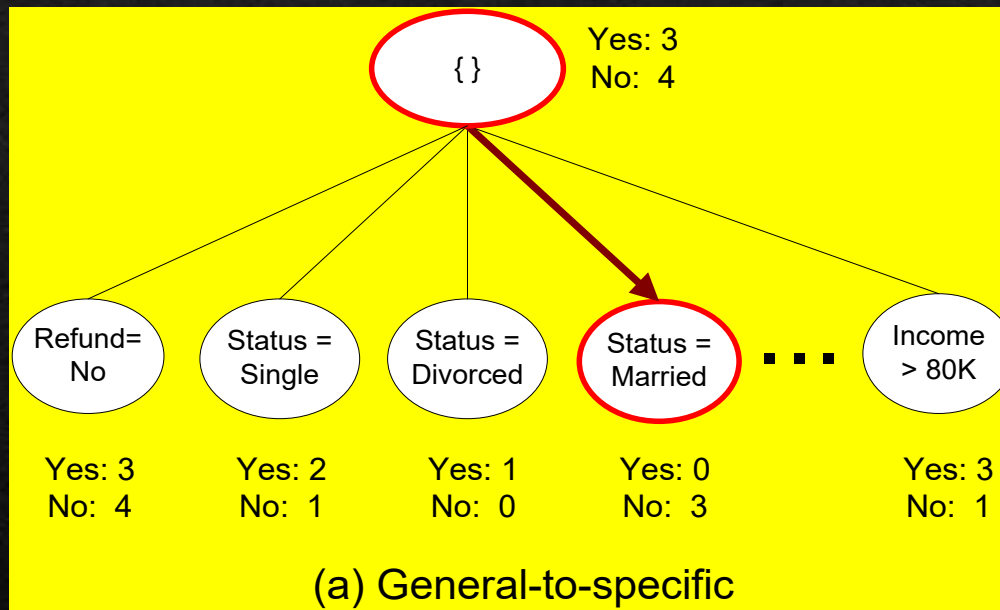
(ii) Step 1

Example of Sequential Covering...



Rule Growing

- Two common strategies



Rule Evaluation

FOIL: First Order Inductive
Learner – an early rule-
based learning algorithm

♦ Foil's Information Gain

- R_0 : $[] \Rightarrow \text{class}$ (initial rule)
- R_1 : $[A] \Rightarrow \text{class}$ (rule after adding conjunct)

$$\text{Gain}(R_0, R_1) = p_1 \times \left[\log_2 \left(\frac{p_1}{p_1 + n_1} \right) - \log_2 \left(\frac{p_0}{p_0 + n_0} \right) \right]$$

- p_0 : number of positive instances covered by R_0
 n_0 : number of negative instances covered by R_0
 p_1 : number of positive instances covered by R_1
 n_1 : number of negative instances covered by R_1

Direct Method: RIPPER

- ◆ For 2-class problem, choose one of the classes as positive class, and the other as negative class
 - Learn rules for positive class
 - Negative class will be default class
- ◆ For multi-class problem
 - Order the classes according to increasing class prevalence (fraction of instances that belong to a particular class)
 - Learn the rule set for smallest class first, treat the rest as negative class
 - Repeat with next smallest class as positive class

Direct Method: RIPPER

- ◆ Growing a rule:
 - Start from empty rule
 - Add conjuncts as long as they improve FOIL's information gain
 - Stop when rule no longer covers negative examples
 - Prune the rule immediately using incremental reduced error pruning
 - Measure for pruning: $v = (p-n)/(p+n)$
 - ◆ p : number of positive examples covered by the rule in the validation set
 - ◆ n : number of negative examples covered by the rule in the validation set
 - Pruning method: delete any final sequence of conditions that maximizes v

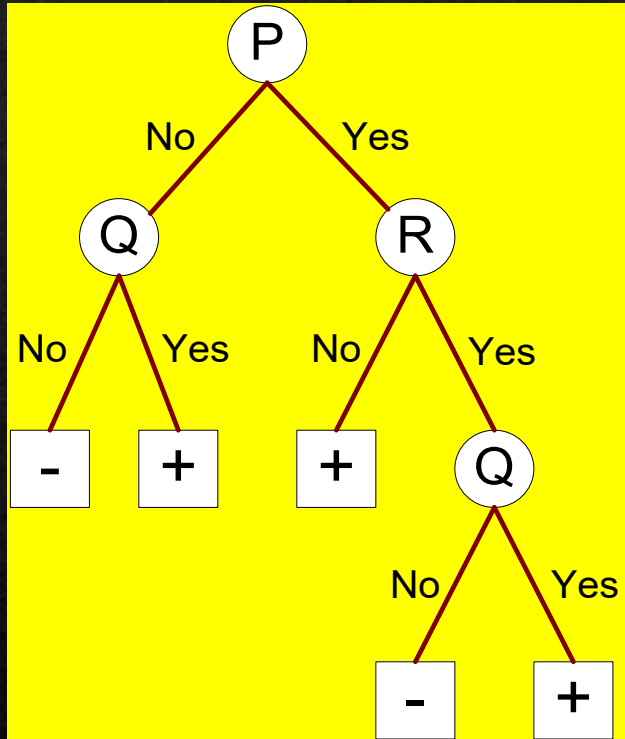
Direct Method: RIPPER

- ◆ Building a Rule Set:
 - Use sequential covering algorithm
 - ◆ Finds the best rule that covers the current set of positive examples
 - ◆ Eliminate both positive and negative examples covered by the rule
 - Each time a rule is added to the rule set, compute the new description length
 - ◆ Stop adding new rules when the new description length is d bits longer than the smallest description length obtained so far

Direct Method: RIPPER

- ◆ Optimize the rule set:
 - For each rule r in the rule set R
 - ◆ Consider 2 alternative rules:
 - Replacement rule (r^*): grow new rule from scratch
 - Revised rule (r'): add conjuncts to extend the rule r
 - ◆ Compare the rule set for r against the rule set for r^* and r'
 - ◆ Choose rule set that minimizes MDL principle
 - Repeat rule generation and rule optimization for the remaining positive examples

Indirect Methods



Rule Set

r1: (P=No,Q=No) ==> -

r2: (P=No,Q=Yes) ==> +

r3: (P=Yes,R=No) ==> +

r4: (P=Yes,R=Yes,Q=No) ==> -

r5: (P=Yes,R=Yes,Q=Yes) ==> +

Indirect Method: C4.5rules

- ◆ Extract rules from an unpruned decision tree
- ◆ For each rule, $r: A \rightarrow y$,
 - consider an alternative rule $r': A' \rightarrow y$ where A' is obtained by removing one of the conjuncts in A
 - Compare the pessimistic error rate for r against all r 's
 - Prune if one of the alternative rules has lower pessimistic error rate
 - Repeat until we can no longer improve generalization error

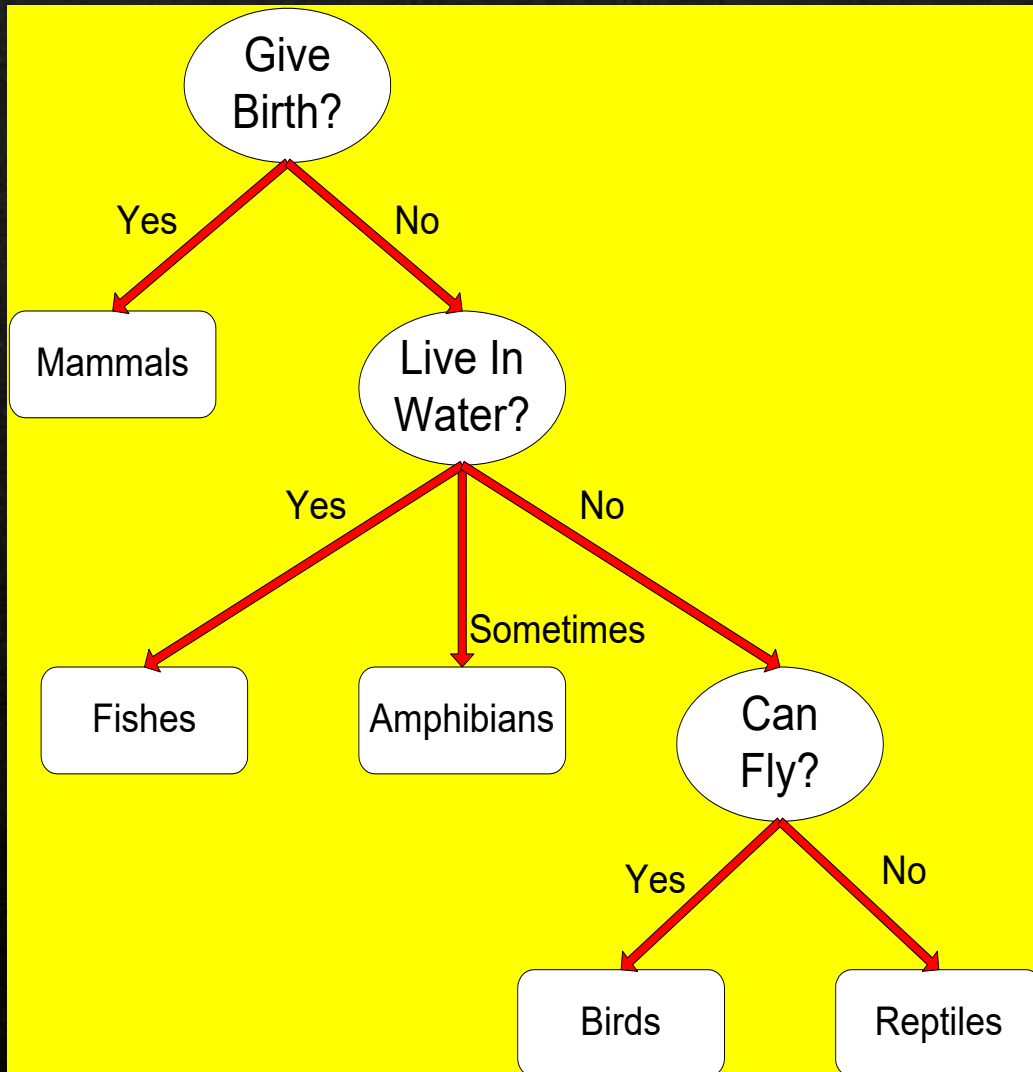
Indirect Method: C4.5 Rules

- ◆ Instead of ordering the rules, order subsets of rules (**class ordering**)
 - Each subset is a collection of rules with the same rule consequent (class)
 - Compute description length of each subset
 - ◆ $\text{Description length} = L(\text{error}) + g L(\text{model})$
 - ◆ g is a parameter that takes into account the presence of redundant attributes in a rule set (default value = 0.5)

Example

Name	Give Birth	Lay Eggs	Can Fly	Live in Water	Have Legs	Class
human	yes	no	no	no	yes	mammals
python	no	yes	no	no	no	reptiles
salmon	no	yes	no	yes	no	fishes
whale	yes	no	no	yes	no	mammals
frog	no	yes	no	sometimes	yes	amphibians
komodo	no	yes	no	no	yes	reptiles
bat	yes	no	yes	no	yes	mammals
pigeon	no	yes	yes	no	yes	birds
cat	yes	no	no	no	yes	mammals
leopard shark	yes	no	no	yes	no	fishes
turtle	no	yes	no	sometimes	yes	reptiles
penguin	no	yes	no	sometimes	yes	birds
porcupine	yes	no	no	no	yes	mammals
eel	no	yes	no	yes	no	fishes
salamander	no	yes	no	sometimes	yes	amphibians
gila monster	no	yes	no	no	yes	reptiles
platypus	no	yes	no	no	yes	mammals
owl	no	yes	yes	no	yes	birds
dolphin	yes	no	no	yes	no	mammals
eagle	no	yes	yes	no	yes	birds

C4.5 versus C4.5rules versus RIPPER



C4.5rules:

(Give Birth=No, Can Fly=Yes) → Birds

(Give Birth=No, Live in Water=Yes) → Fishes

(Give Birth=Yes) → Mammals

(Give Birth=No, Can Fly=No, Live in Water=No) → Reptiles

() → Amphibians

RIPPER:

(Live in Water=Yes) → Fishes

(Have Legs=No) → Reptiles

(Give Birth=No, Can Fly=No, Live In Water=No)
→ Reptiles

(Can Fly=Yes, Give Birth=No) → Birds

() → Mammals

C4.5 versus C4.5rules versus RIPPER

C4.5 and C4.5rules:

		PREDICTED CLASS				
		Amphibians	Fishes	Reptiles	Birds	Mammals
ACTUAL CLASS	Amphibians	2	0	0	0	0
	Fishes	0	2	0	0	1
	Reptiles	1	0	3	0	0
	Birds	1	0	0	3	0
	Mammals	0	0	1	0	6

RIPPER:

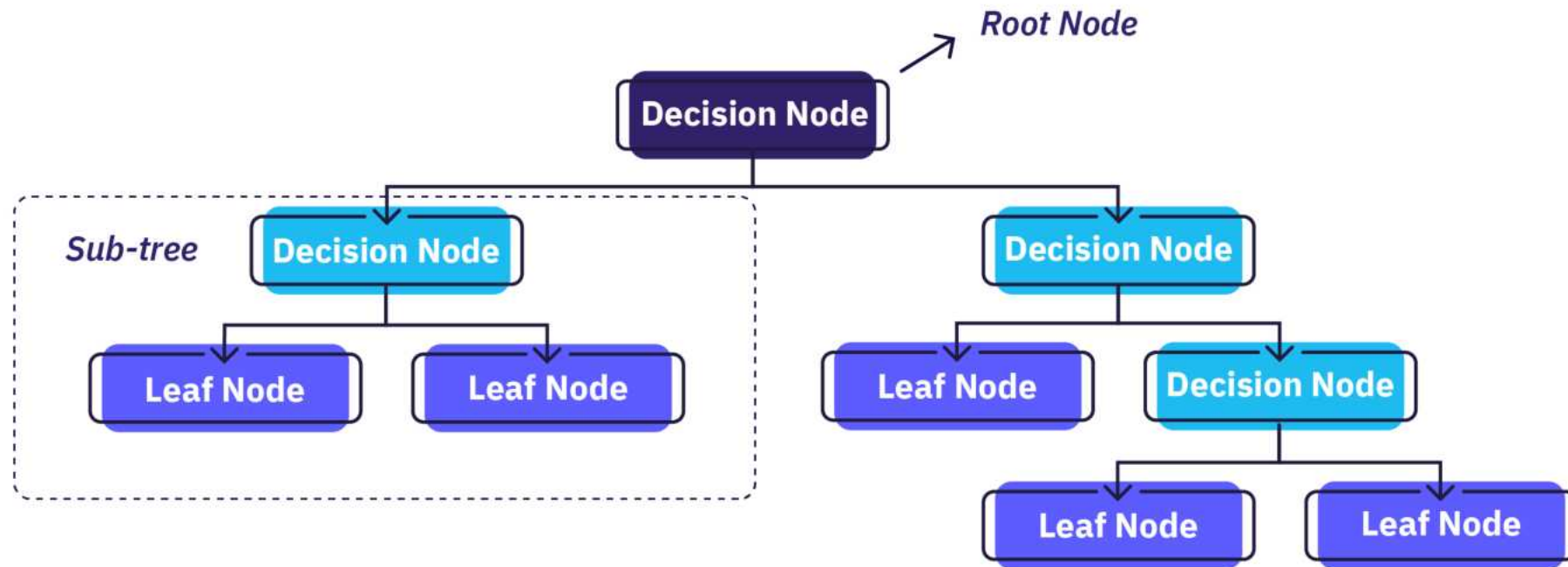
		PREDICTED CLASS				
		Amphibians	Fishes	Reptiles	Birds	Mammals
ACTUAL CLASS	Amphibians	0	0	0	0	2
	Fishes	0	3	0	0	0
	Reptiles	0	0	3	0	1
	Birds	0	0	1	2	1
	Mammals	0	2	1	0	4

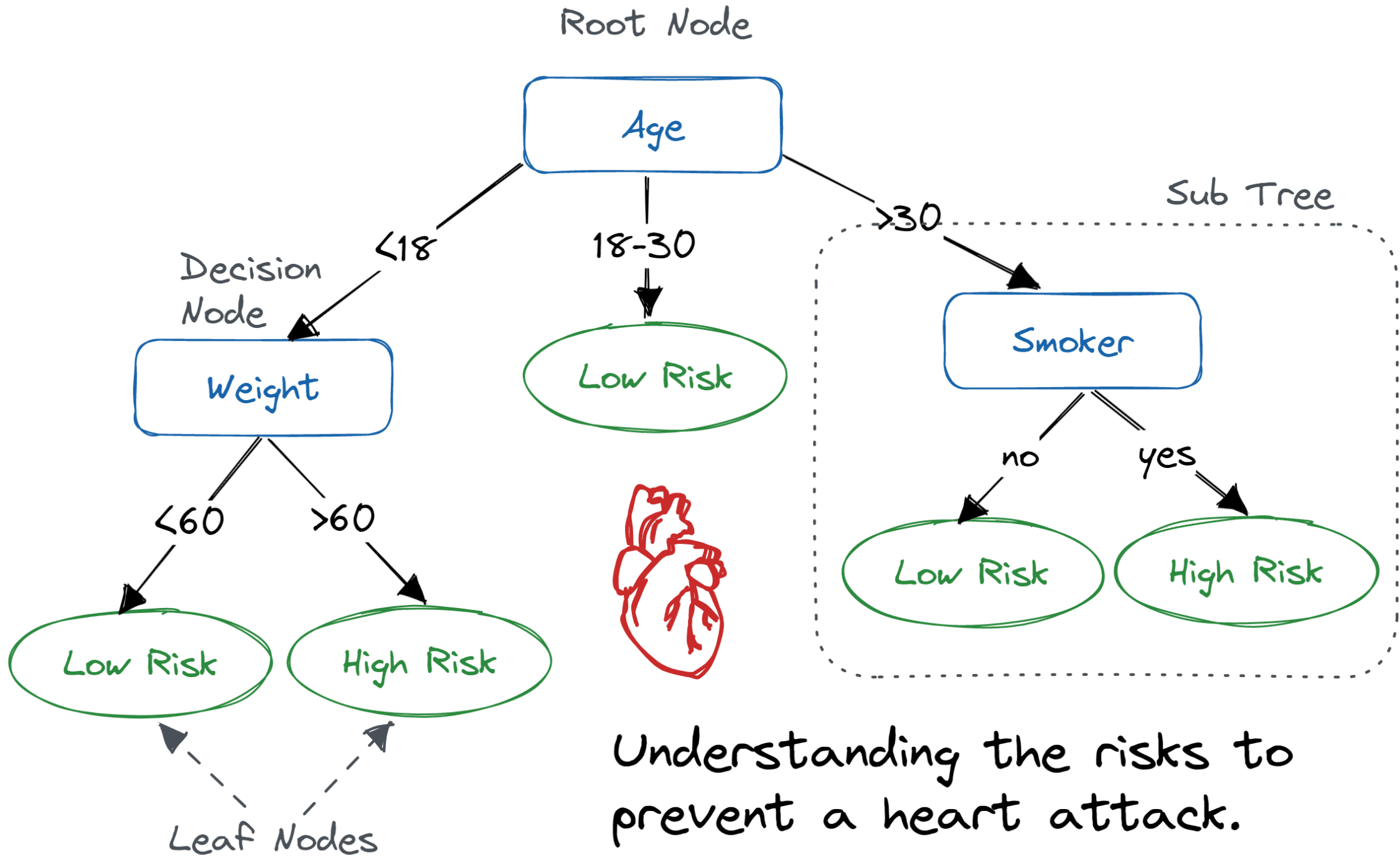
Advantages of Rule-Based Classifiers

- ◆ Has characteristics quite similar to decision trees
 - As highly expressive as decision trees
 - Easy to interpret (if rules are ordered by class)
 - Performance comparable to decision trees
 - ◆ Can handle redundant and irrelevant attributes
 - ◆ Variable interaction can cause issues (e.g., X-OR problem)
- ◆ Better suited for handling imbalanced classes
- ◆ Harder to handle missing values in the test set

Decision Trees

Decision Trees in HR Analytics





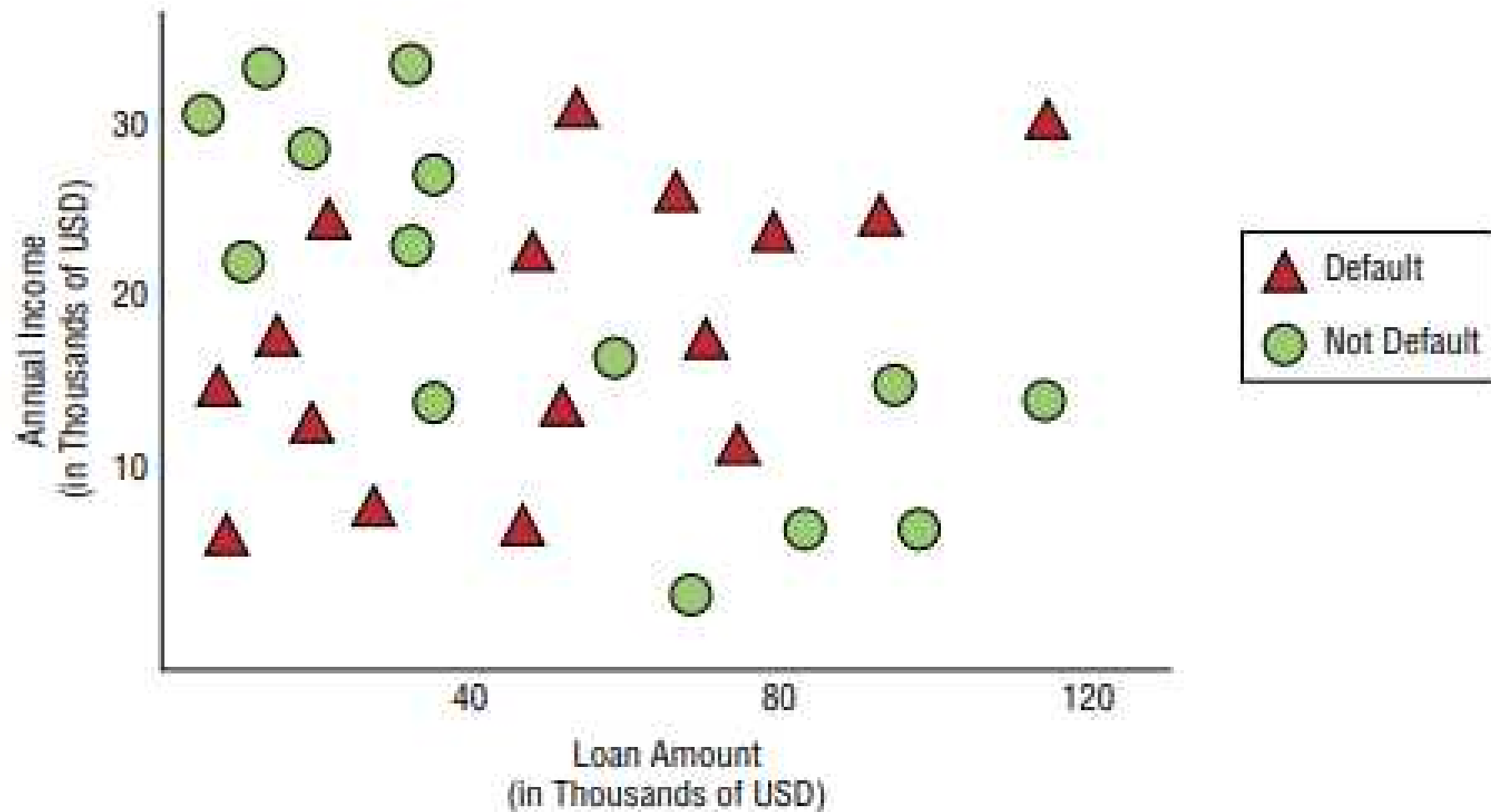


Figure 8.2 Scatterplot of annual income versus loan amount for 30 commercial bank customers (including loan outcomes)

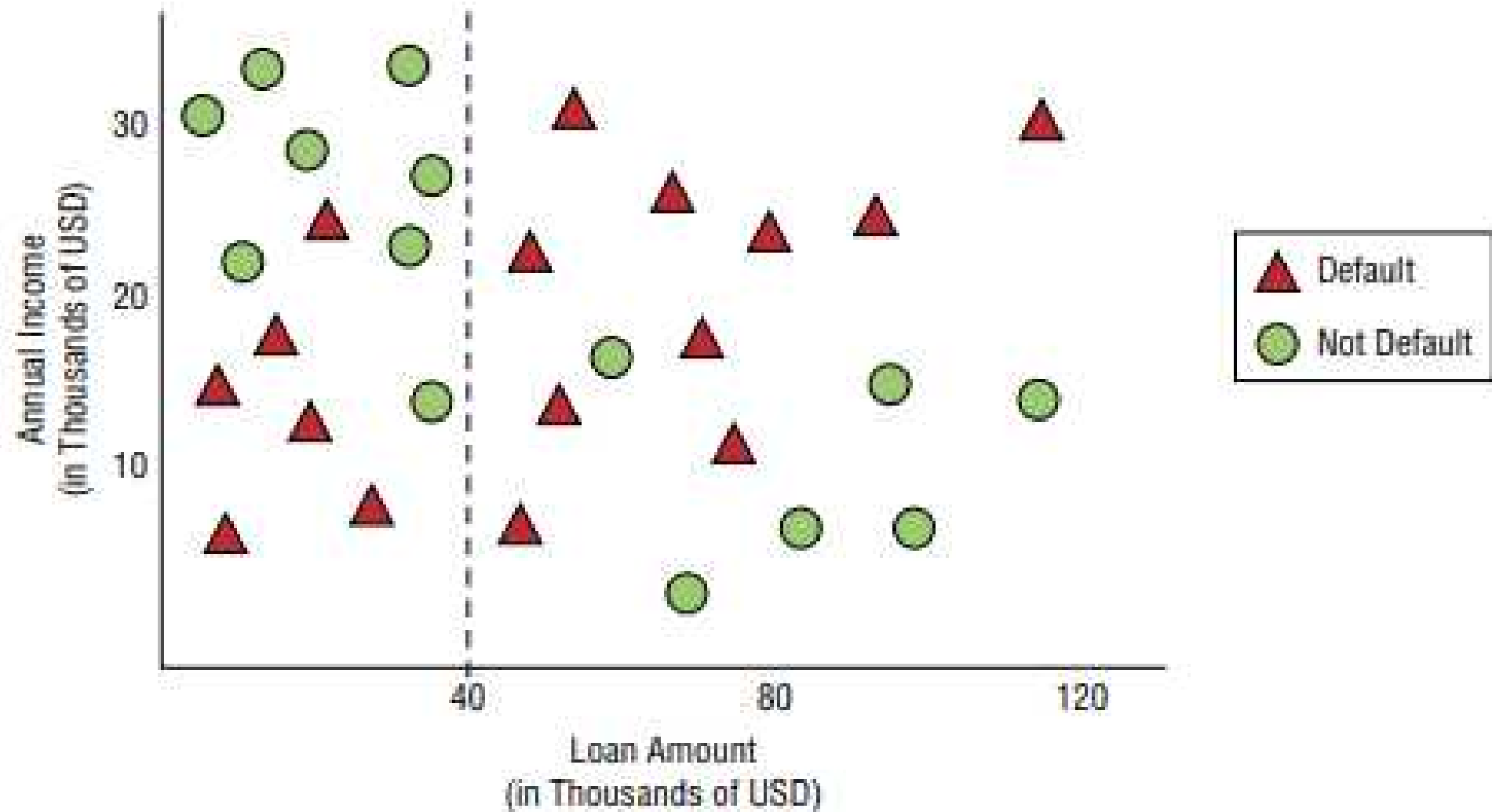


Figure 8.3 Bank customers partitioned on loan amount of less than or more than \$40,000

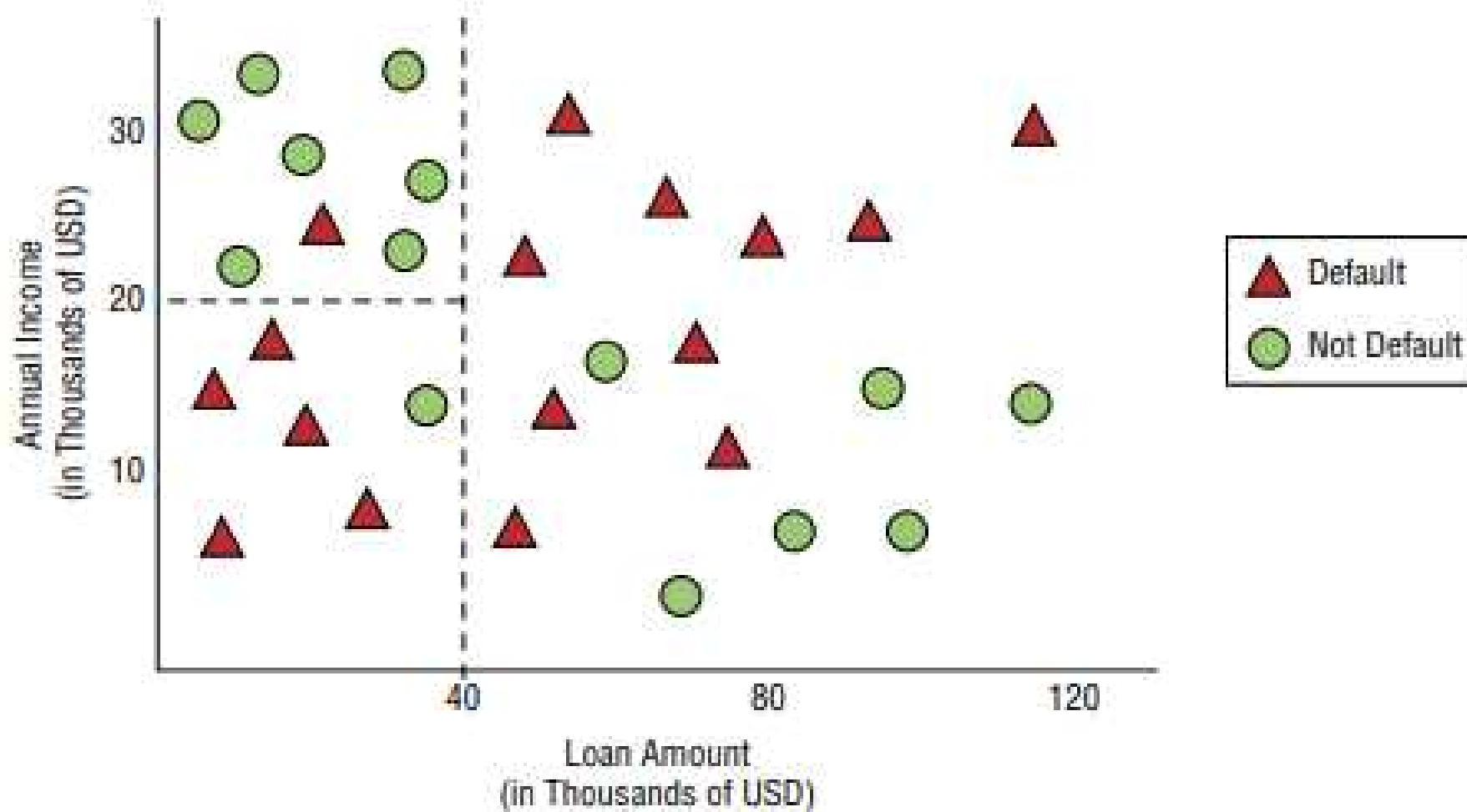


Figure 8.4 Bank customers partitioned on loan amount of less than or more than \$40,000 as well as on annual income less than or more than \$20,000 a year

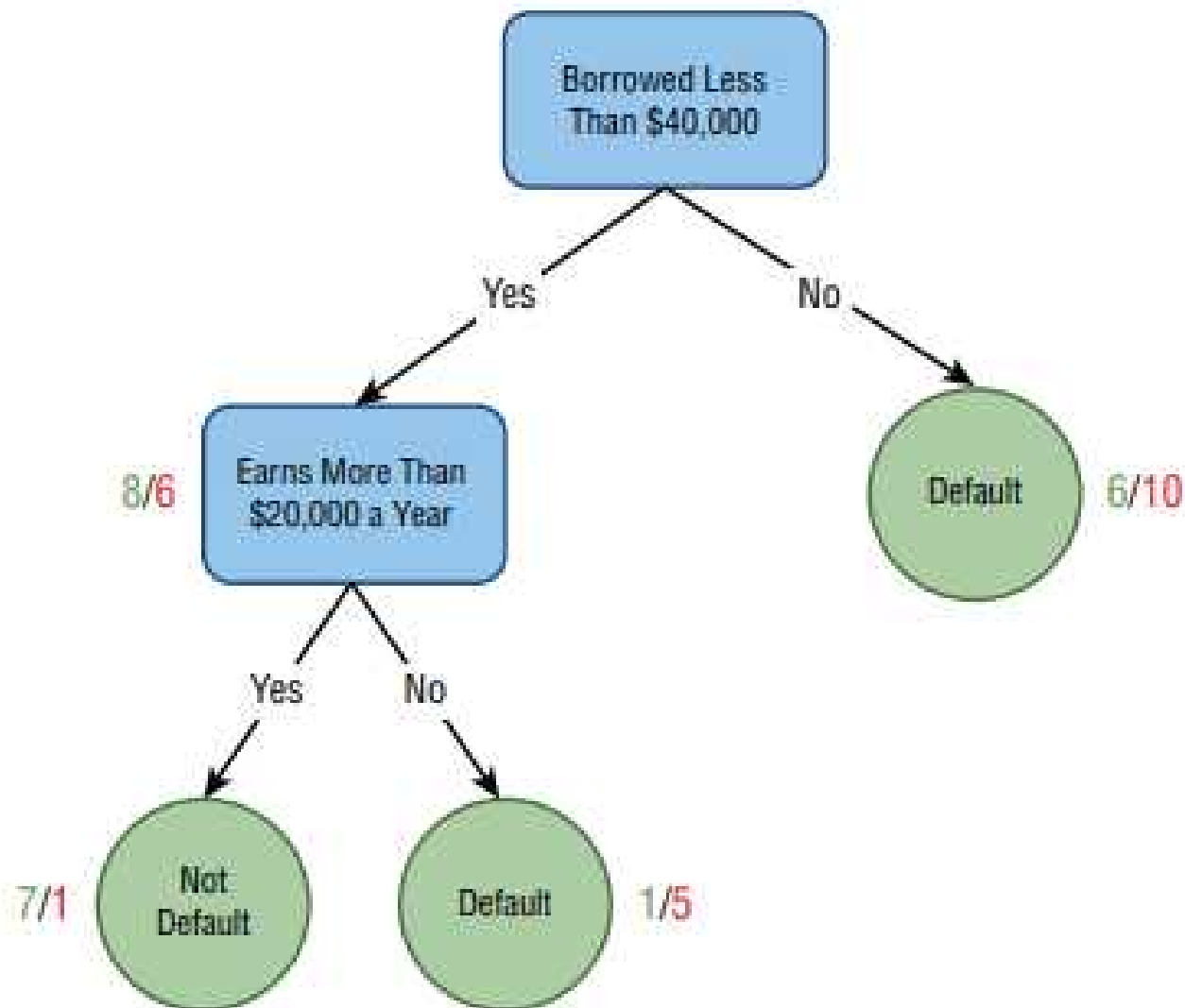


Figure 8.5 Decision tree of bank customers based on the loan amount and annual income. Each decision and leaf node shows the number of customers who defaulted (red number) and those who did not (green number).

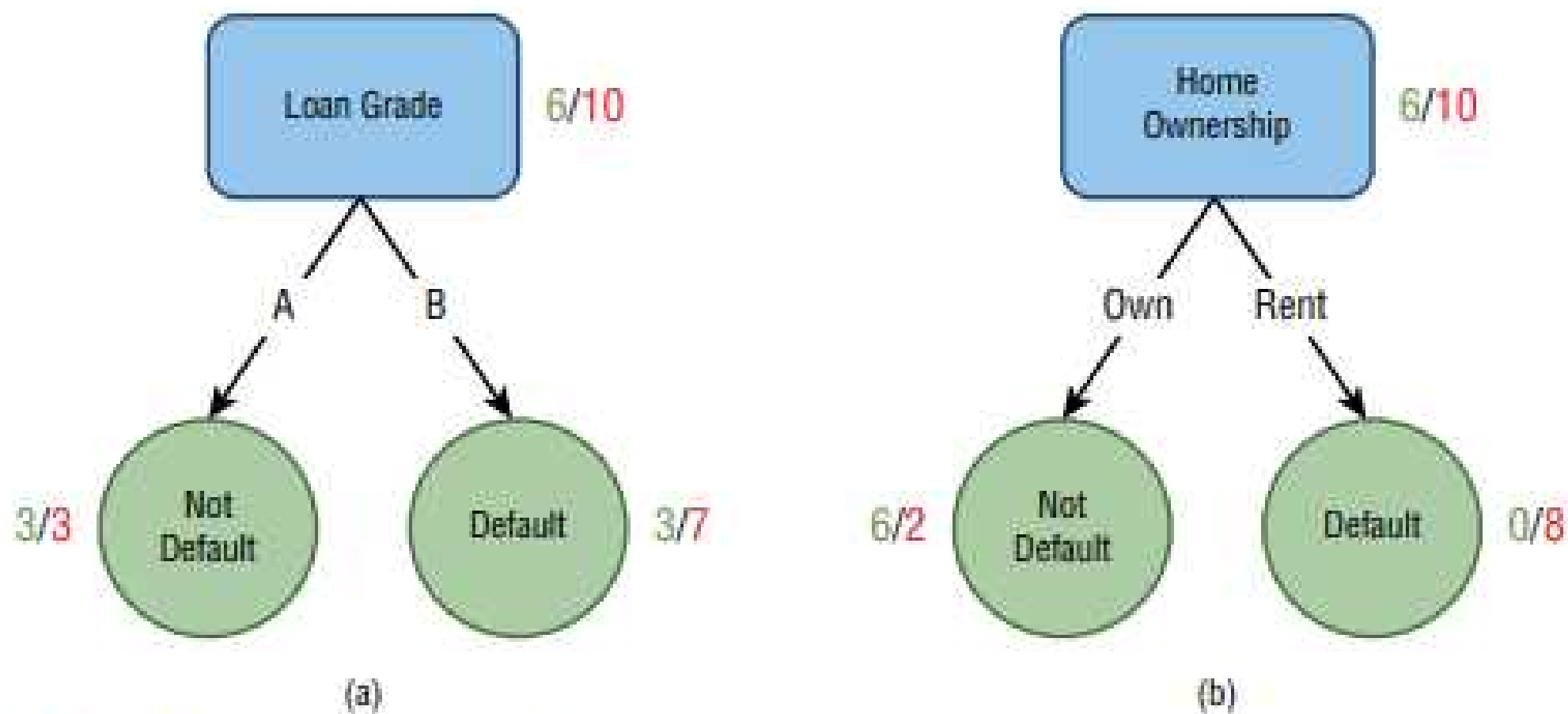


Figure 8.6 Candidate features for splitting the partition of customers who borrowed more than \$40,000. Each decision and leaf node shows the number of customers who defaulted (red number) and those who did not (green number).

The C5.0 Decision Tree Algorithm

◆ Strengths

- ◆ An all-purpose classifier that does well on most problems
- ◆ Highly automatic learning process, which can handle numerical or nominal features
- ◆ Can be used on both small and large datasets

◆ Weakness

- ◆ Decision tree models are often biased toward splits on features having a large number of levels
- ◆ It is easy to overfit or underfit the model
- ◆ Can have trouble modeling some relationships due to reliance on axis-parallel splits

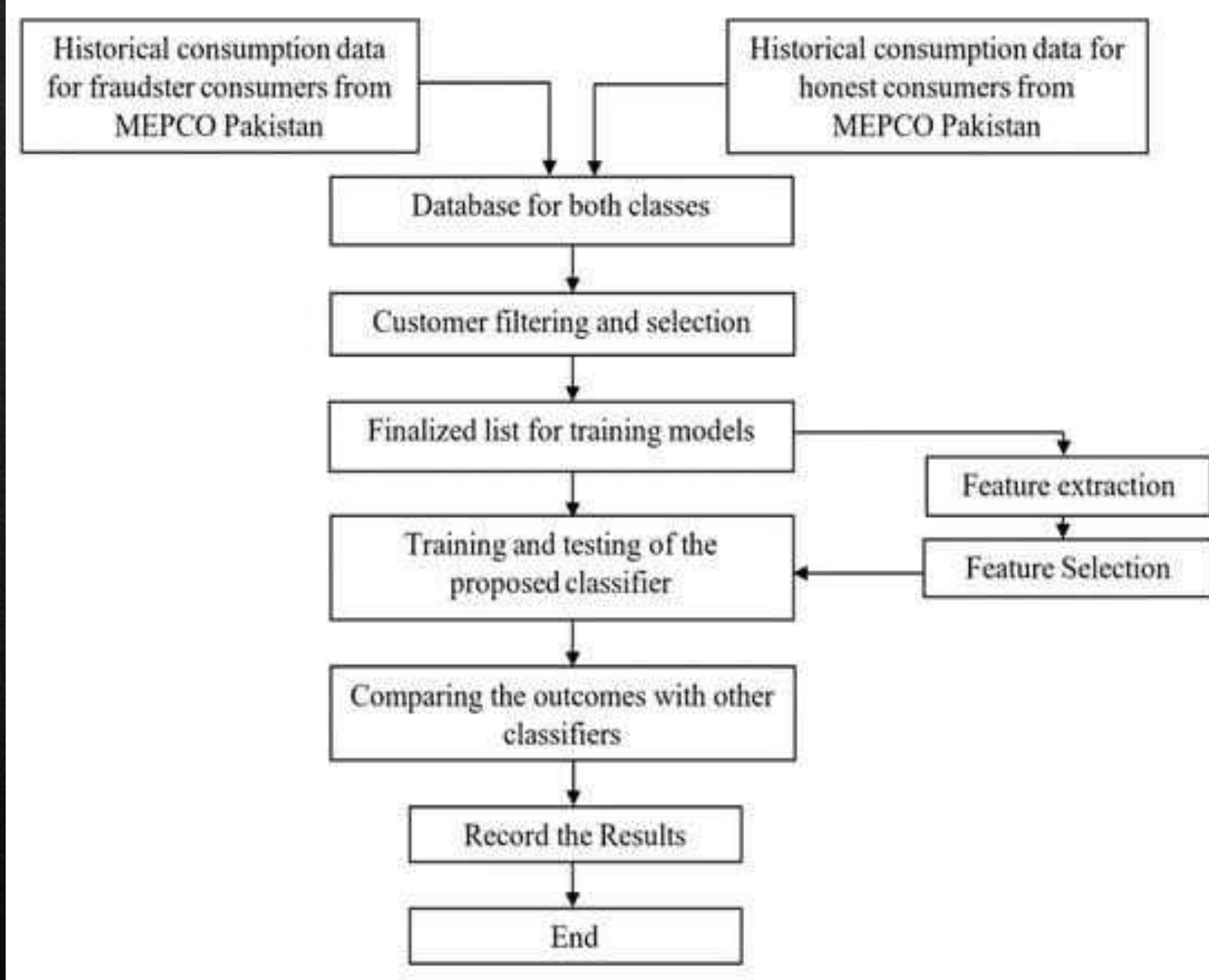
The C5.0 Decision Tree Algorithm

◆ Strengths

- ◆ Results in a model can be interpreted without a mathematical background (for relatively small trees)
- ◆ More efficient than other complex models

◆ Weakness

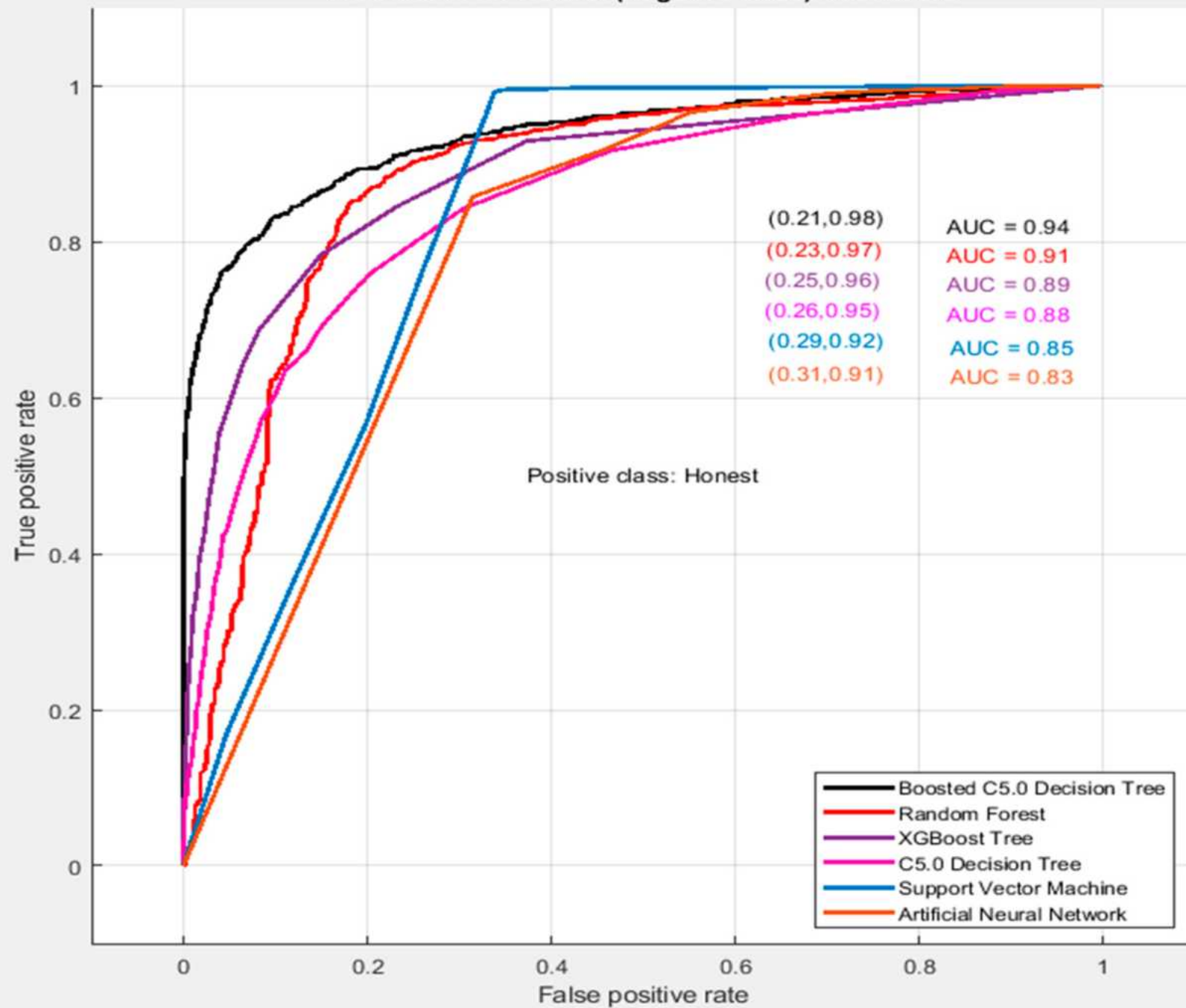
- ◆ Small changes in the training data can result in large changes to decision logic
- ◆ Large trees can be difficult to interpret, and the decisions they make may seem counterintuitive

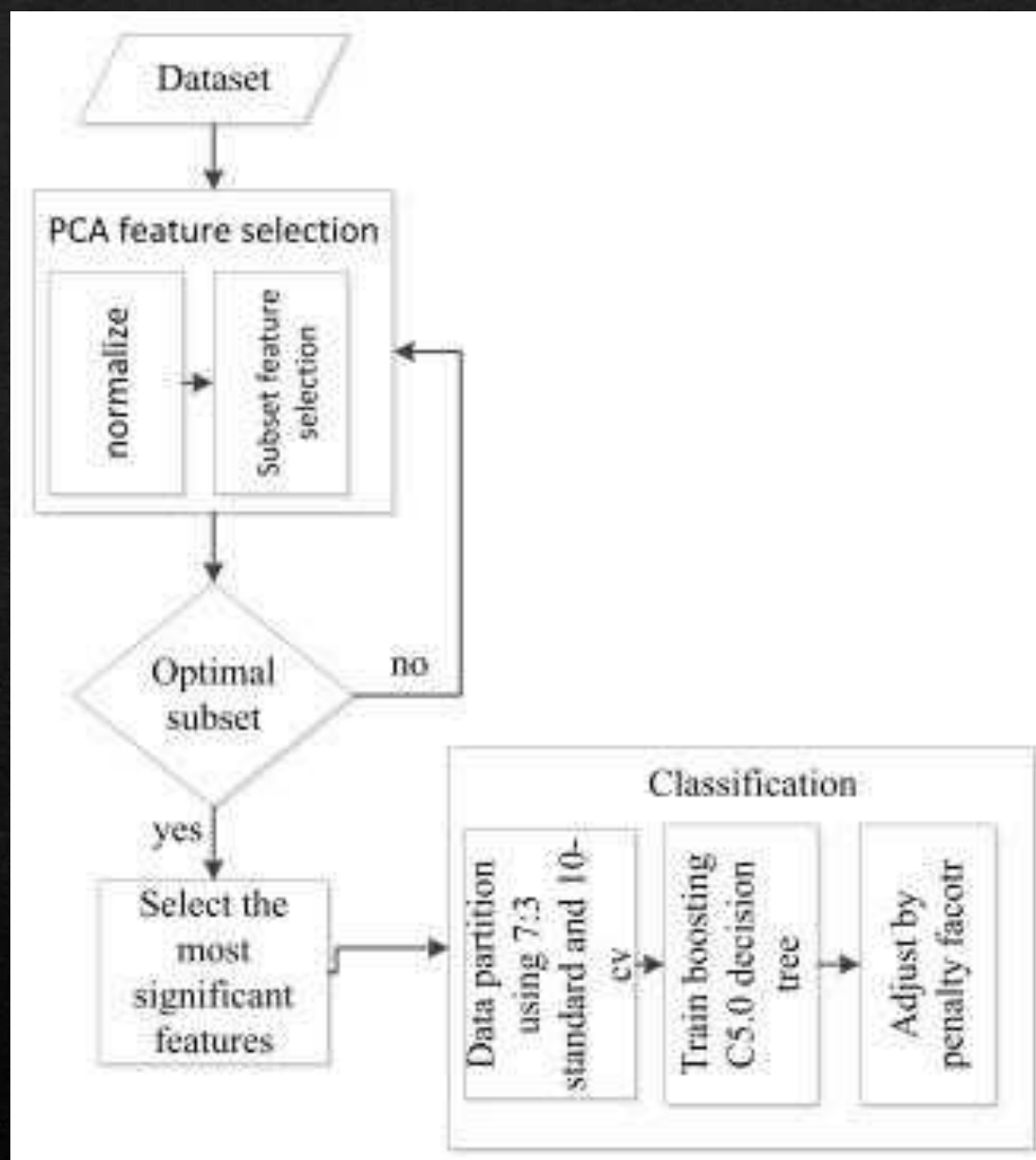


An Efficient Boosted C5.0 Decision-Tree-Based Classification Approach for Detecting Non-Technical Losses in Power Utilities.

Energies **2020**, *13*(12), 3242; <https://doi.org/10.3390/en13123242>

ROC Curve for Honest (Negative class) Customers





Breast cancer diagnosis using feature extraction and boosted C5.0 decision tree algorithm with a penalty factor. Mathematical Biosciences and Engineering, 2022, volume 19, Issue 3: 2193 -2205. doi:10.3934/mbe.2022102